Categories of Boolean networks

A Dissertation
Presented to
the Graduate School of
Clemson University

In Partial Fulfillment
of the Requirements for the Degree
Doctor of Philosophy
Mathematics

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May 2028

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Chapter 1

Introduction

This chapter investigates how maps between Boolean networks can respect both their dynamics and the variable dependencies encoded by their wiring diagrams. Two Boolean networks, f and g, are conjugate if $g = \pi \circ f \circ \pi^{-1}$ for some bijection π of the Boolean cube; so conjugacy defines reasonable notion of isomorphism of Boolean networks. Indeed, the state spaces of two conjugate Boolean networks are isomorphic as direct graphs, but their wiring diagrams need not be isomorphic. In fact, an arbitrary relabeling of states need not preserve changes in a single coordinate. We therefore begin with maps π that preserve the Hamming distance on the Boolean cube, which we call *isometries*. Algebraically, these form a group that is isomorphic to the Weyl group of type B_n—this is the well-known group of signed permutations, isomorphic to the wreath product $C_2 \wr S_n$. It is also the semidirect product $C_2^n \rtimes S_n$, because isometry is a composition of a coordinate permutation and coordinate complementation. Conjugating a Boolean network by one of these isometries induces an isomorphism of the wiring diagrams, and thus this provides a natural notion of equivalence for Boolean networks.

Everything in the previous paragraph is well know. Our main goal in this chapter is to extend these ideas of Boolean networks equivalence to reduction and morphisms of Boolean networks. A *semiconjugacy* from an n-node Boolean network f to an m-node Boolean network g, with $n \geq m$, is a surjective state map h between the phase spaces satisfying $h \circ f = g \circ h$. Intuitively, the image g is a reduction of the Boolean network f, and so a semiconjugacy defines a reasonable notion of *homomorphism* between Boolean networks. In this chapter, we will strengthen this concept and define a *homometry* between Boolean networks. Our goal is to understand which Boolean network homomorphisms reduce the wiring diagram in a structure-preserving manner. The following table summarizes the difference between the different types of structure preserving maps between Boolean networks:

	preserves dynamics	also preserves wiring diagram
$\mathbb{F}_2^n \hookrightarrow \mathbb{F}_2^n$	**isomorphism**	**isometry**
$\mathbb{F}_2^n \twoheadrightarrow \mathbb{F}_2^m$	**homomorphism**	**homometry**

Initially, it should not be obvious how to properly define the concept of homometry, or how to characterize these maps. For the first point, we will define that $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$, sending $f \mapsto g$ is a homometry if and only if it is a (weak) *contraction*, i.e., that

$$|h(x) - h(y)| \leq |x - y|, \qquad \text{for all } x, y \in \mathbb{F}_2^n.$$

Then we will prove (see Theorem 2.2.11) that an equivalence condition is that the inverse image of every edge in the wiring diagram of g lifts to an edge in the wiring diagram of f. This leads to the concept of a *node map* between the wiring diagrams. In some sense, this says that structure-

preserving maps between Boolean networks are characterized by the inverse images of a key feature (variable dependencies) being preserved is analogous how continuous maps pull back open sets to open sets, or how group homomorphisms pull back normal subgroups to normal subgroups, or how linear maps pull back subspaces to subspaces. In all of these cases, there is a well-established categorical foundation, leading to the categories of **Top**, **Grp**, and $\mathbf{Vect}_{\mathbb{F}}$, respectively. In this thesis, we will develop the theory of structure-preserving maps between Boolean works, both as a Boolean analogue of continuous maps, and also in a categorical framework, where we explore ideas such as products, coproducts, adjoints, universal properties, and so on.

Chapter 2

Quotients of Boolean Networks

In this chapter, we will review conjugacy and isometric conjugacy of Boolean networks, and extend them to semiconjugacy and homometry, respectively. We focus on surjective state maps and determine which additional condition gives a consistent assignment of source nodes to target nodes. The next chapter will allow maps that are not surjective and study their factorizations through their images.

2.1 Isomorphism and isometry of Boolean networks

The following definition captures precisely the concept of two Boolean networks having isomorphic phase spaces. It need not preserve their wiring diagrams, which record additional information about coordinate dependence. We will strengthen conjugacy by requiring the state map to preserve Hamming distance. This condition ensures that the wiring diagrams are isomorphic, and the networks are then called *isometrically conjugate*. Later in the chapter, we will allow surjective maps between networks of different sizes.

Definition 2.1.1. Let $f, g\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ be Boolean networks. A conjugacy from f to g is a bijection $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ such that $h \circ f = g \circ h$. When such a map exists, we say that f and g are conjugate and write $f \sim g$.

We can represent the previous identity as the following commutative diagram.

$$\begin{array}{ccc} \mathbb{F}_2^n & \xrightarrow{\quad h \quad} & \mathbb{F}_2^n \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ \mathbb{F}_2^n & \xrightarrow{\quad h \quad} & \mathbb{F}_2^n \end{array}$$

Consequently, one may either (i) update a state via f and subsequently relabel it through h, or (ii) first apply the relabeling h and then update via g; in both cases the resulting state is identical. Furthermore, given a Boolean network f and a bijection h, there exists a unique map g that renders the corresponding diagram commutative, namely

$$g = h \circ f \circ h^{-1}.$$

Observe that the identity map conjugates f to itself. If $h \circ f = g \circ h$, then composing on the right with h^{-1} yields $f \circ h^{-1} = h^{-1} \circ g$, and hence h^{-1} conjugates g to f. Finally, if $h \circ f = g \circ h$

and $k \circ g = l \circ k$, then

$$(k \circ h) \circ f \;=\; k \circ (g \circ h) \;=\; l \circ (k \circ h),$$

which shows that $k \circ h$ conjugates f to l. Therefore,

Proposition 2.1.2. *Conjugacy is an equivalence relation on Boolean networks.*

Example 2.1.3. A paradigmatic instance of a structural correspondence is provided by the equivalence between OR-networks and AND-networks. Specifically, consider the maps $f, g\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ defined by

$$f(x_1, x_2, x_3) = (x_1x_2x_3,\ x_1x_2,\ x_1x_3); \quad g(y_1, y_2, y_3) = (y_1 \vee y_2 \vee y_3,\ y_1 \vee y_2,\ y_1 \vee y_3).$$

The wiring diagrams associated with these two networks coincide; they are given below, where self-loops are understood and each edge is regarded as bidirectional:



Define $h\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ by $h(x_1, x_2, x_3) = (\overline{x}_1, \overline{x}_2, \overline{x}_3)$. This map is a bijection (indeed, an involution). Moreover, a direct computation yields

$$\begin{aligned} hf(x_1, x_2, x_3) &= h(x_1x_2x_3,\ x_1x_2,\ x_1x_3) \\ &= (\overline{x}_1 \vee \overline{x}_2 \vee \overline{x}_3,\ \overline{x}_1 \vee \overline{x}_2,\ \overline{x}_1 \vee \overline{x}_3) \\ &= g(\overline{x}_1, \overline{x}_2, \overline{x}_3) \\ &= gh(x_1, x_2, x_3), \end{aligned}$$

and hence $hf = gh$, establishing that f and g are conjugate via h.

It should be observed that the conjugacy identity is invariant under arbitrary iteration. More precisely, for every $k \in \mathbb{N}$,

$$h \circ f^k = g^k \circ h.$$

Here, f^k (respectively, g^k) denotes the k-fold iterate of f (respectively, g), i.e., the k-fold self-composition. It follows that h maps the entire forward orbit of x,

$$x,\ f(x),\ f^2(x), \ldots,$$

onto the forward orbit of $h(x)$,

$$h(x),\ g(h(x)),\ g^2(h(x)), \ldots,$$

in a term-wise manner. Equivalently, in graph-theoretic language, a conjugacy is precisely an isomorphism of directed graphs between the corresponding phase spaces.

Conjugacy exhibits numerous desirable properties that are compatible with dynamical behavior; we have mentioned only a subset of these. Many such properties are investigated in [5, 6, 8, 4], among other works. However, some limitations have been found regarding the geometry of the Boolean networks; we will see this explicitly in the following example.

Example 2.1.4. Let $f, g\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ be the Boolean networks defined by

$$f(x_0, x_1, x_2) = (x_0, 0, 0)$$

and

$$g(x_0, x_1, x_2) = \big(p(x), p(x), p(x)\big), \qquad p(x) = x_0 + x_1 + x_2,$$

where $x = (x_0, x_1, x_2)$ and $+$ denotes addition modulo 2. Define a map $\pi\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ by

$$\pi(x_0, x_1, x_2) = \big(x_0 + x_1,\ x_0 + x_2,\ x_0 + x_1 + x_2\big).$$

To verify that π is bijective, define

$$\pi^{-1}(y_0, y_1, y_2) = \big(y_0 + y_1 + y_2,\ y_1 + y_2,\ y_0 + y_2\big).$$

Using the identities $a + a = 0$ and $a + 0 = a$, together with the associativity and commutativity of $+$, we obtain

$$\begin{aligned}\pi^{-1}(\pi(x_0, x_1, x_2)) &= \Big((x_0 + x_1) + (x_0 + x_2) + (x_0 + x_1 + x_2),\\ &\qquad\qquad (x_0 + x_2) + (x_0 + x_1 + x_2),\\ &\qquad\qquad (x_0 + x_1) + (x_0 + x_1 + x_2)\Big)\\ &= (x_0, x_1, x_2).\end{aligned}$$

Conversely, if

$$(x_0, x_1, x_2) = \pi^{-1}(y_0, y_1, y_2) = \big(y_0 + y_1 + y_2,\ y_1 + y_2,\ y_0 + y_2\big),$$

then

$$\begin{aligned}\pi(\pi^{-1}(y_0, y_1, y_2)) &= \Big((y_0 + y_1 + y_2) + (y_1 + y_2),\\ &\qquad\qquad (y_0 + y_1 + y_2) + (y_0 + y_2),\\ &\qquad\qquad (y_0 + y_1 + y_2) + (y_1 + y_2) + (y_0 + y_2)\Big)\\ &= (y_0, y_1, y_2).\end{aligned}$$

Thus π^{-1} is the inverse of π, so π is a permutation of the eight-element set $\mathbb{F}_2^3$.

For every $(x_0, x_1, x_2) \in \mathbb{F}_2^3$,

$$\pi(f(x_0, x_1, x_2)) = \pi(x_0, 0, 0) = (x_0, x_0, x_0).$$

Now let $(y_0, y_1, y_2) = \pi(x_0, x_1, x_2)$. Then

$$\begin{aligned}y_0 + y_1 + y_2 &= (x_0 + x_1) + (x_0 + x_2) + (x_0 + x_1 + x_2)\\ &= x_0.\end{aligned}$$

It follows that

$$g(\pi(x_0, x_1, x_2)) = g(y_0, y_1, y_2) = (x_0, x_0, x_0).$$

Therefore,

$$g \circ \pi = \pi \circ f.$$

The corresponding wiring diagrams of f and g are as follows:



The only essential dependence in f is that of f_0 on x_0, whereas each coordinate function of g depends essentially on all three variables. Hence,

$$|E(W(f))| = 1, \qquad |E(W(g))| = 9.$$

Since an isomorphism of directed graphs preserves the number of edges,

$$W(f) \ncong W(g).$$

A general theory of the above example is explored in [7].

The preceding example shows that the phase space alone does not determine the wiring diagram. The distinction arises because a wiring diagram records the effect of changing one coordinate at a time, whereas an arbitrary conjugacy need not preserve such changes. In fact, the map π in Example 2.1.4 sends the states $(0,0,0)$ and $(1,0,0)$, which differ in only one coordinate, to $(0,0,0)$ and $(1,1,1)$, which differ in all three coordinates. To obtain a notion of conjugacy that also preserves the wiring diagram, we therefore seek relabelings that respect the geometry of the Boolean cube.

2.1.1 The Hamming distance and the Boolean cube

Definition 2.1.5. For $x, y \in \mathbb{F}_2^n$, the *Hamming distance* between x and y is

$$d_H(x,y) = \big|\{i \in \{1,\dots,n\} : x_i \neq y_i\}\big|.$$

The *Boolean cube* Q_n is the undirected graph with vertex set $\mathbb{F}_2^n$, in which two states are adjacent if their Hamming distance is 1.

The Hamming distance is a metric. Indeed, it is nonnegative and symmetric, and $d_H(x,y) = 0$ if and only if $x = y$. For the triangle inequality, observe that if $x_i \neq z_i$, then either $x_i \neq y_i$ or $y_i \neq z_i$. Counting these coordinates gives

$$d_H(x,z) \leq d_H(x,y) + d_H(y,z).$$

Moreover, $d_H(x,y)$ is precisely the length of a shortest path from x to y in Q_n: every differing coordinate must be changed at least once, and changing each of them exactly once provides such a path. Thus Q_n is the graph of vertices and edges of the unit cube $[0,1]^n$. Its edges describe single-coordinate changes, independently of any Boolean network defined on its vertices.

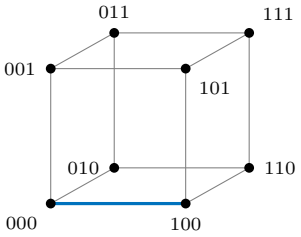
Example 2.1.6. In $\mathbb{F}_2^3$, we have

$$d_H\big((0,0,0),(1,0,0)\big) = 1, \qquad d_H\big((0,0,0),(1,1,1)\big) = 3.$$

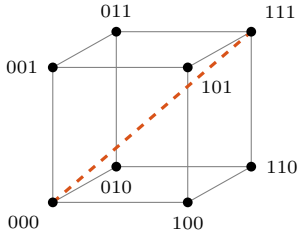
Returning to the conjugacy π in Example 2.1.4, these equalities show that

$$d_H\big(\pi(0,0,0),\pi(1,0,0)\big) \neq d_H\big((0,0,0),(1,0,0)\big).$$

The highlighted edge on the left is sent to the pair of opposite vertices on the right. The dashed segment is a body diagonal of the unit cube, not an edge of Q_3.



$d_H(000,100) = 1$



$d_H\big(\pi(000),\pi(100)\big) = 3$

Here and in subsequent cube drawings, a string such as 100 denotes the state $(1,0,0)$. The two cubes above display the selected pair and its image under π; their remaining vertices retain their usual coordinate labels.

Definition 2.1.7. A map $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ is an *isometry* of the Boolean cube if

$$d_H\big(h(x),h(y)\big) = d_H(x,y) \qquad \text{for all } x,y \in \mathbb{F}_2^n.$$

A Boolean network is called *isometric* if its update map is an isometry.

Every isometry is injective, since $h(x) = h(y)$ implies $d_H(x,y) = 0$, and hence $x = y$. Since $\mathbb{F}_2^n$ is finite, every isometry of the Boolean cube is therefore bijective. Its inverse is also an isometry. Equivalently, an isometry is an automorphism of Q_n: distance preservation gives adjacency preservation in both directions, and a graph automorphism preserves shortest-path lengths.

Definition 2.1.8. Let $f, g\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ be Boolean networks. An *isometric conjugacy* from f to g is an isometry $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ such that

$$h \circ f = g \circ h.$$

When such a map exists, we say that f and g are *isometrically conjugate* and write $f \sim_{\text{iso}} g$.

In this definition, the distance-preserving condition is imposed on h; the update maps f and g need not themselves be isometries. Since the identity, inverses, and compositions of isometries are isometries, the same argument used for conjugacy shows that isometric conjugacy is an equivalence relation on Boolean networks.

Example 2.1.9. Consider the map $h\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ defined by

$$h(x_1,x_2,x_3) = (\overline{x}_3, x_1, \overline{x}_2).$$

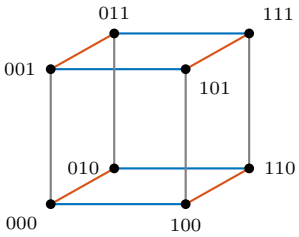
Complementing a coordinate does not change whether two entries agree, and permuting the coordinates does not change the number of disagreements. Consequently,

$$\begin{aligned} d_H\big(h(x),h(y)\big) &= |\overline{x}_3 - \overline{y}_3| + |x_1 - y_1| + |\overline{x}_2 - \overline{y}_2| \\ &= |x_3 - y_3| + |x_1 - y_1| + |x_2 - y_2| \\ &= d_H(x,y), \end{aligned}$$

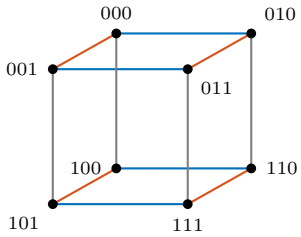
where the differences in this calculation are taken in $\mathbb{R}$. Thus h is an isometry, with inverse

$$h^{-1}(y_1, y_2, y_3) = (y_2, \overline{y}_3, \overline{y}_1).$$

The following drawings show the action of h on every vertex of the unit cube. Vertices in corresponding positions are related by h. The blue, orange, and gray edges represent changes in x_1, x_2, x_3, respectively; their images represent changes in y_2, y_3, y_1.



$x = (x_1, x_2, x_3)$



$y = h(x)$

The complementation map from Example 2.1.3 is also an isometry. Hence the AND-network and OR-network in that example are isometrically conjugate. Neither update map is an isometry: the AND-network sends both $(0,0,0)$ and $(0,0,1)$ to $(0,0,0)$, while the OR-network sends both $(1,0,0)$ and $(1,1,0)$ to $(1,1,1)$.

2.1.2 Preservation of the wiring diagram

We now show that isometric conjugacy preserves the wiring diagram up to a relabeling of its vertices. Let $e_i \in \mathbb{F}_2^n$ denote the vector whose ith coordinate is 1 and whose other coordinates are 0. We use $+$ componentwise on vectors, so $x + e_i$ is the state obtained from x by changing its ith coordinate. Recall that the wiring diagram $W(f)$ has vertex set $\{1, \ldots, n\}$, with

$$i \longrightarrow j \text{ in } W(f) \quad \Longleftrightarrow \quad f_j(x) \neq f_j(x + e_i) \text{ for some } x \in \mathbb{F}_2^n. \tag{2.1}$$

This is the unsigned wiring diagram; self-loops are included. Its vertices are coordinates, whereas the vertices of Q_n are states.

The essential geometric point is that an isometry sends all edges in a given coordinate direction to edges in one fixed coordinate direction. The choice of the new direction cannot depend on the state.

Lemma 2.1.10. *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ be an isometry. There exists a unique permutation σ of $\{1, \ldots, n\}$ such that*

$$h(x + e_i) = h(x) + e_{\sigma(i)} \qquad \text{for every } x \in \mathbb{F}_2^n \text{ and every } i. \tag{2.2}$$

Moreover, for all $u, v \in \mathbb{F}_2^n$,

$$h(u)_{\sigma(j)} \neq h(v)_{\sigma(j)} \quad \Longleftrightarrow \quad u_j \neq v_j. \tag{2.3}$$

Proof. Since h is an automorphism of Q_n, it maps the n neighbors of each x bijectively onto the n neighbors of $h(x)$. Thus there is a permutation σ_x such that

$$h(x + e_i) = h(x) + e_{\sigma_x(i)}.$$

Traversing the same edge in the opposite direction shows that $\sigma_{x+e_i}(i) = \sigma_x(i)$.

Now let $i \neq j$. The four distinct states

$$x, \quad x + e_i, \quad x + e_i + e_j, \quad x + e_j$$

form a square in Q_n. Put $z = h(x)$, $r = \sigma_x(i)$, and $s = \sigma_x(j)$. The two common neighbors of $z + e_r$ and $z + e_s$ are precisely z and $z + e_r + e_s$. Since h is injective, the vertex opposite x must therefore satisfy

$$h(x + e_i + e_j) = z + e_r + e_s = h(x + e_j) + e_r.$$

Therefore, $\sigma_{x+e_j}(i) = \sigma_x(i)$. Consequently, $\sigma_x(i)$ is unchanged when any coordinate of x is changed. The cube is connected, so σ_x is a single permutation σ, independent of x. Its uniqueness follows from the images of the edges incident with $\mathbf{0}$.

Finally, join u to v by changing each coordinate in which they differ exactly once. Equation (2.2) shows that the image path changes precisely the corresponding coordinates $\sigma(i)$, again exactly once. In particular, its endpoints differ in coordinate $\sigma(j)$ if and only if u and v differ in coordinate j, proving equation (2.3). □

Proposition 2.1.11. *Let $f, g\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ be isometrically conjugate via h, and let σ be the permutation associated with h in Lemma 2.1.10. Then*

$$i \longrightarrow j \text{ in } W(f) \quad \Longleftrightarrow \quad \sigma(i) \longrightarrow \sigma(j) \text{ in } W(g).$$

Thus $\sigma\colon W(f) \to W(g)$ is an isomorphism of directed graphs.

Proof. Fix $x \in \mathbb{F}_2^n$ and put $y = h(x)$. By equation (2.2),

$$h(x + e_i) = y + e_{\sigma(i)}.$$

The conjugacy identity gives

$$g(y) = h(f(x)), \qquad g(y + e_{\sigma(i)}) = h(f(x + e_i)).$$

Applying equation (2.3) to these two output states yields

$$g_{\sigma(j)}(y) \neq g_{\sigma(j)}(y + e_{\sigma(i)}) \quad \Longleftrightarrow \quad f_j(x) \neq f_j(x + e_i).$$

Since h is bijective, y ranges over all of $\mathbb{F}_2^n$ as x does. Taking the existence of such a state on both sides and using equation (2.1) proves the assertion, including the case $i = j$. □

Example 2.1.12. Consider the Boolean network

$$f(x_1, x_2, x_3) = (x_1x_2,\ x_1 \vee x_3,\ x_1 + x_2)$$

and the isometry $h(x_1, x_2, x_3) = (\overline{x}_3, x_1, \overline{x}_2)$ from Example 2.1.9. Define $g = h \circ f \circ h^{-1}$. Substituting the expression for h^{-1} gives

$$\begin{aligned} g(y_1, y_2, y_3) &= h\big(y_2\overline{y}_3,\ y_2 \vee \overline{y}_1,\ y_2 + \overline{y}_3\big) \\ &= \big(y_2 + y_3,\ y_2\overline{y}_3,\ y_1\overline{y}_2\big). \end{aligned}$$

For example, at $x = (1, 0, 1)$,

$$f(x) = (0, 1, 1), \qquad h(f(x)) = (0, 0, 0),$$

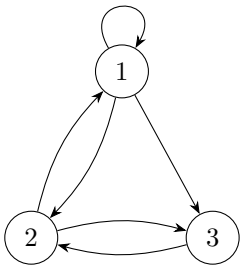
while $h(x) = (0, 1, 1)$ and $g(h(x)) = (0, 0, 0)$, as required by the conjugacy identity.

The coordinate directions are permuted according to

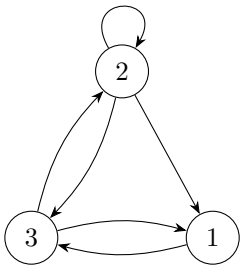
$$\sigma(1) = 2, \qquad \sigma(2) = 3, \qquad \sigma(3) = 1.$$

The wiring diagrams are drawn below, with corresponding vertices placed in corresponding positions.

Every arrow on the left has its image on the right; in particular, the self-loop at 1 becomes the self-loop at 2.



$W(f)$



$W(g)$

The corresponding map of unit cubes is exactly the one displayed in Example 2.1.9. The wiring diagrams have only three vertices, one for each coordinate, while each cube has eight vertices, one for each state. In this example, $W(f)$ and $W(g)$ are isomorphic but are not equal as labeled graphs: their self-loops have different labels.

Remark 2.1.13. Proposition 2.1.11 concerns unsigned wiring diagrams. Complementing coordinates can change whether a dependence is increasing or decreasing. For example, the first coordinate function $f_1 = x_1x_2$ in Example 2.1.12 is increasing in x_2, whereas the corresponding function $g_2 = y_2\overline{y}_3$ is decreasing in y_3. Thus preservation of the unsigned diagram does not assert preservation of individual edge signs.

2.1.3 Characterization of the isometries

Example 2.1.9 suggests that coordinate permutations and coordinate complementations are the basic isometries of the Boolean cube. We now show that every isometry is obtained in this way. This standard characterization is also recorded in [2].

For a permutation σ of $\{1, \ldots, n\}$, let $P_\sigma \colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ be the coordinate permutation defined by

$$(P_\sigma(x))_{\sigma(i)} = x_i, \qquad \text{equivalently} \qquad (P_\sigma(x))_j = x_{\sigma^{-1}(j)}.$$

Thus an input coordinate i is moved to output coordinate $\sigma(i)$, consistently with the relabeling of the wiring diagram above. For $a \in \mathbb{F}_2^n$, define $\tau_a(x) = a + x$. This map complements precisely those coordinates j for which $a_j = 1$.

Theorem 2.1.14. *A map $h \colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ is an isometry if and only if there exist $a \in \mathbb{F}_2^n$ and a permutation σ of $\{1, \ldots, n\}$ such that*

$$h = \tau_a \circ P_\sigma.$$

The pair (a, σ) is unique, with $a = h(\mathbf{0})$ and

$$h(e_i) + h(\mathbf{0}) = e_{\sigma(i)}.$$

Proof. Suppose first that h is an isometry. Put $a = h(\mathbf{0})$ and let σ be the permutation from Lemma 2.1.10. Starting at $\mathbf{0}$ and changing each coordinate i for which $x_i = 1$, equation (2.2) gives

$$h(x) = a + \sum_{\{i : x_i = 1\}} e_{\sigma(i)} = a + P_\sigma(x).$$

Here the empty sum is $\mathbf{0}$. Evaluating at $\mathbf{0}$ determines a, and evaluating at each e_i then determines σ, proving uniqueness.

Conversely, suppose that $h(x) = a + P_\sigma(x)$. For every $x, y \in \mathbb{F}_2^n$,

$$\begin{aligned} d_H\big(h(x), h(y)\big) &= \big|\{j : a_j + x_{\sigma^{-1}(j)} \neq a_j + y_{\sigma^{-1}(j)}\}\big| \\ &= \big|\{j : x_{\sigma^{-1}(j)} \neq y_{\sigma^{-1}(j)}\}\big| \\ &= d_H(x, y). \end{aligned}$$

Hence h is an isometry. □

Consequently, there are exactly $2^n n!$ isometries of Q_n, and an isometry is completely determined by its values at $\mathbf{0}, e_1, \ldots, e_n$. In particular, an isometry fixing $\mathbf{0}$ is a coordinate permutation. The characterization also gives the coordinate form of an isometric conjugacy: if $g = h \circ f \circ h^{-1}$ and $h = \tau_a \circ P_\sigma$, then

$$g_j(y) = a_j + f_{\sigma^{-1}(j)}\big(P_{\sigma^{-1}}(y + a)\big).$$

When σ is the identity, the wiring diagram is unchanged as a labeled graph. For a general σ, its vertices are relabeled by σ.

Example 2.1.15. For $h(x_1, x_2, x_3) = (\overline{x}_3, x_1, \overline{x}_2)$, the images of the origin and its three neighbors are

$$h(000) = 101, \qquad h(100) = 111, \qquad h(010) = 100, \qquad h(001) = 001.$$

Thus $a = (1, 0, 1)$, and

$$h(e_1) + a = e_2, \qquad h(e_2) + a = e_3, \qquad h(e_3) + a = e_1.$$

We recover $\sigma = (123)$ and

$$h(x) = (1, 0, 1) + (x_3, x_1, x_2),$$

in agreement with the two labeled cubes in Example 2.1.9. These four images determine the images of all eight states. By contrast, the map π from Example 2.1.4 fixes 000 but sends 100 to 111; it therefore cannot be a coordinate permutation and hence cannot be an isometry. Proposition 2.1.11 further shows that no other conjugacy between the two networks in Example 2.1.4 can be isometric, since their wiring diagrams are not isomorphic.

The characterization has a particularly simple interpretation when the update map of a Boolean network is itself an isometry.

Corollary 2.1.16. *A Boolean network $f\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ is isometric if and only if every vertex of $W(f)$ has exactly one incoming edge and exactly one outgoing edge. Equivalently, $W(f)$ is a disjoint union of directed cycles, with self-loops allowed as cycles of length one.*

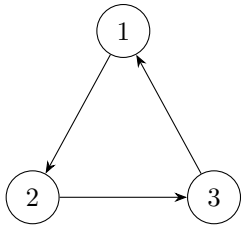
Proof. If f is isometric, Theorem 2.1.14 gives $f_{\sigma(i)}(x) = a_{\sigma(i)} + x_i$. Its wiring diagram therefore has exactly the edges $i \longrightarrow \sigma(i)$, so every vertex has one incoming and one outgoing edge.

Conversely, suppose that every vertex has one incoming and one outgoing edge, and write its unique outgoing edge as $i \longrightarrow \sigma(i)$. The incoming-edge condition makes σ a permutation. The function $f_{\sigma(i)}$ depends essentially only on x_i: changing any other coordinate does not change its value, and any two states with the same ith coordinate can be joined by such changes. A nonconstant Boolean function of one variable is either that variable or its complement. Hence $f_{\sigma(i)}(x) = a_{\sigma(i)} + x_i$ for some $a_{\sigma(i)} \in \mathbb{F}_2$, and Theorem 2.1.14 applies. □

Example 2.1.17. Regard the isometry from Example 2.1.9 as an update map, and denote the resulting network by

$$q(x_1, x_2, x_3) = (\overline{x}_3, x_1, \overline{x}_2).$$

Its wiring diagram and all of its periodic orbits are given below.



$W(q)$

Fixed points: 001 and 110.

$000 \longrightarrow 101 \longrightarrow 011 \longrightarrow 000,$

$100 \longrightarrow 111 \longrightarrow 010 \longrightarrow 100.$

The corresponding cube map appears in Example 2.1.9. The directed cycle in $W(q)$ records coordinate dependencies, whereas the two cycles of length three above record the evolution of states. Although q preserves distances between pairs of updated states, it need not move each state along a cube edge: $d_H(000, q(000)) = 2$.

2.2 Homomorphism and homometry of Boolean networks

Conjugacy describes a change of labels on the state space. To compare a network with a smaller system that records only some of its information, we must also allow distinct states to have the same image. We therefore replace the bijective state map by a surjective one. The dynamical condition remains a commutative square, but its consequences for the Boolean cube and the wiring diagram require separate consideration.

Throughout this section, $n, m \geq 1$, $[n] = \{1, \ldots, n\}$, and $+$ denotes addition modulo 2. We use the Hamming distance on cubes of either dimension, and e_i denotes the standard basis vector in the cube indicated by its context.

2.2.1 Semiconjugacy and the identification of states

Definition 2.2.1. Let $f\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ and $g\colon \mathbb{F}_2^m \to \mathbb{F}_2^m$ be Boolean networks. A *semiconjugacy*, or *homomorphism*, from f to g is a surjective Boolean map $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ such that

$$h \circ f = g \circ h. \tag{2.4}$$

Thus the following diagram commutes:

$$\begin{array}{ccc} \mathbb{F}_2^n & \xrightarrow{\quad h \quad} & \mathbb{F}_2^m \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ \mathbb{F}_2^n & \xrightarrow[\quad h \quad]{} & \mathbb{F}_2^m \end{array}$$

When such a map exists, g is called a *factor* of f. Surjectivity is part of our convention for semiconjugacy.

The map h records an observation of a state. Equation (2.4) says that this observation can be updated using g without knowing which state in the fiber $h^{-1}(y)$ produced it. Surjectivity ensures that every state of the smaller network is represented. It also implies $m \leq n$. If $m = n$, then h is bijective and the semiconjugacy is a conjugacy.

As with conjugacy, induction gives $h \circ f^k = g^k \circ h$ for every $k \geq 0$. Thus h maps each forward orbit onto the corresponding forward orbit in the factor. Distinct trajectories can merge under h, and the period of a periodic state can decrease to a divisor of its original period. Identity maps are semiconjugacies, and compositions of semiconjugacies are semiconjugacies.

Proposition 2.2.2. *Let $f\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ and let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be surjective. There is a Boolean network g satisfying equation (2.4) if and only if*

$$h(x) = h(x') \quad \Longrightarrow \quad h(f(x)) = h(f(x')) \qquad (x, x' \in \mathbb{F}_2^n). \tag{2.5}$$

If g exists, it is unique.

Proof. Necessity follows by applying g to $h(x) = h(x')$. Conversely, for $y \in \mathbb{F}_2^m$, choose $x \in h^{-1}(y)$ and put $g(y) = h(f(x))$. The condition in equation (2.5) makes this definition independent of the choice of x. It gives $gh = hf$, and surjectivity of h forces these values of g at every y. □

Example 2.2.3. Consider the eight-node network

$$f(x) = \big(x_2, x_3, x_4, \overline{x}_1, x_1 + x_5, x_2x_6, x_3 \vee x_7, \overline{x}_8\big)$$

and the four-node network

$$g(y) = \big(y_2, y_3, y_4, \overline{y}_1\big).$$

The projection $h(x) = (x_1, x_2, x_3, x_4)$ is surjective, and

$$h(f(x)) = (x_2, x_3, x_4, \overline{x}_1) = g(h(x)).$$

Every state y has 16 preimages: its first four coordinates are prescribed, and the last four are arbitrary. Although the discarded coordinates have their own dynamics and receive inputs from retained coordinates, they do not affect the evolution of the observation.

This reduction cannot preserve all distances. For instance, $h(\mathbf{0}) = h(e_5)$ even though $d_H(\mathbf{0}, e_5) = 1$. It does satisfy

$$d_H(h(x), h(x')) = \big|\{i \le 4 : x_i \ne x'_i\}\big| \le d_H(x, x').$$

Thus information can be discarded without increasing the size of a perturbation.

Example 2.2.4. Semiconjugacy alone does not impose the preceding inequality. In fact, its failure can be arbitrarily large. Let $m \ge 2$ and $r \ge 1$, write a source state as $(u, z) \in \mathbb{F}_2^m \times \mathbb{F}_2^r$, and let $R\colon \mathbb{F}_2^r \to \mathbb{F}_2^r$ cyclically permute the coordinates. Define

$$\begin{aligned} h(u, z) &= (u_1, u_1 + u_2, \ldots, u_1 + u_m), \\ f(u, z) &= (\overline{u}_1, 0, \ldots, 0; R(z)), \\ g(y) &= (\overline{y}_1, \overline{y}_1, \ldots, \overline{y}_1). \end{aligned}$$

Here the semicolon separates the first m updated coordinates from the last r. Given y, take $u_1 = y_1$ and $u_i = y_1 + y_i$ for $2 \le i \le m$; any z then gives a preimage. Moreover,

$$h(f(u, z)) = (\overline{u}_1, \ldots, \overline{u}_1) = g(h(u, z)).$$

These are nonconstant dynamics, and g exchanges $\mathbf{0}$ and $\mathbf{1}$. Nevertheless,

$$d_H(\mathbf{0}, e_1) = 1, \qquad d_H(h(\mathbf{0}), h(e_1)) = m.$$

One changed input can therefore appear as a simultaneous change in every observed coordinate. The case $m = 5$, $r = 3$ already gives an eight-node network mapping onto a five-node network with this behavior.

2.2.2 Contractions of Boolean cubes

Distance preservation would force h to be injective, excluding every proper reduction of the state space. Example 2.2.3 suggests allowing distances to decrease, while Example 2.2.4 explains why an upper bound is an additional condition.

Definition 2.2.5. A Boolean map $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ is a *contraction* if

$$d_H(h(x), h(x')) \leq d_H(x, x') \qquad \text{for all } x, x' \in \mathbb{F}_2^n. \tag{2.6}$$

A semiconjugacy satisfying this condition is a *contractive semiconjugacy*, also called a *homometry*.

The inequality in equation (2.6) is usually called *Hamming nonexpansiveness*, or the 1-Lipschitz condition; see, for example, [9]. We use *contraction* for this non-strict inequality. A bound with a constant strictly smaller than 1 would identify the endpoints of every cube edge, and connectedness of the cube would force the map to be constant. The contraction condition here concerns h; the update maps f and g need not be contractions.

Lemma 2.2.6. *A map $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ is a contraction if and only if*

$$d_H(h(x + e_i), h(x)) \leq 1 \qquad \textit{for every } x \in \mathbb{F}_2^n \textit{ and } i \in [n]. \tag{2.7}$$

Equivalently, each edge of Q_n maps either to an edge of Q_m or to a single vertex.

Proof. Necessity follows by taking two inputs at distance 1. Conversely, join x to x' by a path of length $d_H(x, x')$ in Q_n. Each step changes the image by distance at most 1. The triangle inequality bounds the distance between the image endpoints by the length of this path. □

For Boolean self-maps, this criterion says that changing one input affects at most one output at any fixed state. It is the local out-degree characterization of nonexpansiveness in [9, Proposition 2]. The proof above applies also when the two cubes have different dimensions. Compositions of contractions are contractions. A surjective contraction from $\mathbb{F}_2^n$ to itself is an isometry. Indeed, it is a bijection of a finite set, so $h^N = \mathrm{Id}$ for some $N \geq 1$. Repeated application of equation (2.6) gives

$$d_H(x, x') \geq d_H(h(x), h(x')) \geq \cdots \geq d_H(h^N(x), h^N(x')) = d_H(x, x'),$$

so all inequalities are equalities.

The edge criterion leaves open which output coordinate changes. Unlike Lemma 2.1.10, it does not assert that this coordinate is determined by the input direction alone. The following example shows both this obstruction and its effect on wiring diagrams.

Example 2.2.7. For $u \in \mathbb{F}_2^3$, let $|u|$ be its number of nonzero coordinates, and define $H\colon \mathbb{F}_2^3 \to \mathbb{F}_2^2$ by

$\lvert u\rvert$	0	1	2	3
$H(u)$	00	10	11	01

The four possible outputs occur, so H is surjective. Adjacent input states have consecutive weights, and consecutive entries in the second row are adjacent in Q_2. Lemma 2.2.6 therefore shows that H is a contraction.

Consider the networks

$$F(u_1, u_2, u_3) = (\overline{u}_3, \overline{u}_1, \overline{u}_2), \qquad G(v_1, v_2) = (v_1, \overline{v}_2).$$

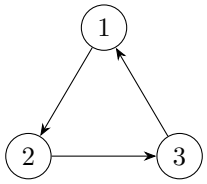
Since $|F(u)| = 3 - |u|$, the table gives $H(F(u)) = G(H(u))$. Thus H is a contractive semiconjugacy between two nonidentity networks. Nevertheless, two edges in input direction 1 have images in

different directions:

$$000 \longleftrightarrow 100 \quad \overset{H}{\longmapsto} \quad 00 \longleftrightarrow 10,$$
$$010 \longleftrightarrow 110 \quad \overset{H}{\longmapsto} \quad 10 \longleftrightarrow 11.$$

Hence no single output direction can be assigned to this input direction at every state.

There is also a direct obstruction to lifting wiring edges. The diagrams are



$W(F)$



$W(G)$

Every surjection $[3] \to [2]$ has a singleton fiber. The loop at its image node would have to lift to a loop at that single source node, but $W(F)$ has no loops. Thus contraction and semiconjugacy do not even guarantee the existence of a surjective node map lifting every target edge.

This obstruction persists in larger systems with interacting additional coordinates. For $r \geq 2$, let R cyclically permute r coordinates and set

$$h(u, z) = (H(u), z), \qquad f(u, z) = (F(u), R(z)), \qquad g(v, z) = (G(v), R(z)).$$

These give a contractive semiconjugacy from $r+3$ nodes to $r+2$ nodes. The source wiring diagram is the disjoint union of a directed 3-cycle and a directed r-cycle, so it has no loops. The target has two loops and a directed r-cycle. Any surjection $[r+3] \to [r+2]$ has exactly one fiber of size 2 and all remaining fibers of size 1. Both loop vertices would need a fiber of size at least 2, which is impossible. Taking $r = 4$ gives a seven-node to six-node example, and r can be arbitrarily large.

2.2.3 Compatible node maps and aggregation

To connect a state map to wiring diagrams, we need a consistent assignment of input coordinates to output coordinates. We allow an input perturbation to disappear, and we allow several inputs to affect the same output. We require that an input cannot affect different outputs at different states.

For example, consider

$$h(x_1, \ldots, x_9) = \big(x_1 \vee x_2 \vee x_3,\ x_4x_5x_6,\ x_7 + x_8 + x_9\big).$$

Changing any of the first three inputs affects only the first output, and the other two groups behave similarly. Some changes disappear: changing x_1 has no effect when $x_2 = 1$. Nevertheless, there is an unambiguous assignment of all nine source nodes to three target nodes. Each aggregation function is nonconstant, and its inputs are disjoint from the inputs of the others, so every output state can be attained independently. This example suggests the following definition.

Definition 2.2.8. Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be a surjective Boolean map. A *compatible node map* for h is a surjection $\phi\colon [n] \to [m]$ such that

$$h(x + e_i) + h(x) \in \{\mathbf{0}, e_{\phi(i)}\} \qquad \text{for every } x \in \mathbb{F}_2^n \text{ and } i \in [n]. \tag{2.8}$$

We write $B_a = \phi^{-1}(a)$ for its *node fibers*. Thus the sets B_a form a partition of $[n]$ into nonempty blocks.

The maps $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ and $\phi\colon [n] \to [m]$ act on different sets: h identifies states, while ϕ groups nodes. In particular, a node fiber B_a is not a state fiber $h^{-1}(y)$. No condition on the edges of $W(f)$ or $W(g)$ is included in Definition 2.2.8; the edge-lifting conclusion will be deduced from compatibility and semiconjugacy.

Proposition 2.2.9. *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be surjective and let $\phi\colon [n] \to [m]$ be a surjection. The following conditions are equivalent:*

(i) ϕ is compatible with h.

(ii) For every $a \in [m]$, the function h_a depends only on the coordinates in $B_a = \phi^{-1}(a)$.

(iii) There are nonconstant Boolean functions $H_a\colon \mathbb{F}_2^{B_a} \to \mathbb{F}_2$ such that

$$h(x) = \big(H_1(x|_{B_1}), \dots, H_m(x|_{B_m})\big). \tag{2.9}$$

Here $\mathbb{F}_2^{B_a}$ denotes states indexed by the nodes in B_a, and $x|_{B_a}$ is the restriction of x to those coordinates. These conditions imply that h is a contraction.

Proof. Suppose (i) holds. If $i \notin B_a$, equation (2.8) gives $h_a(x+e_i) = h_a(x)$. Any two states agreeing on B_a can be joined by changing only coordinates outside B_a, so they have the same value of h_a. This proves (ii). Condition (ii) defines H_a by restriction, and surjectivity of h ensures that each h_a, hence each H_a, takes both Boolean values. Thus (iii) holds.

Conversely, under (iii), changing an input coordinate i can affect only the output with $a = \phi(i)$, which gives (i). Finally, equation (2.8) implies equation (2.7), so Lemma 2.2.6 gives contraction. □

This proposition identifies the extra restriction beyond contraction. At a fixed state, a contraction permits an input to affect at most one output; compatibility requires the same assignment at every state. Equivalently, a surjective h admits a compatible node map precisely when each input coordinate is essential in at most one of the functions $h_1, \dots, h_m$. An input essential in h_a must be assigned to a. Inputs ignored by every component of h can be assigned to any node. Each output has an essential input because it is nonconstant, so these assignments give a surjection. When a compatible node map exists, it is unique exactly when every input is used or $m = 1$.

The projection in Example 2.2.3 therefore has compatible node maps: the first four nodes have forced assignments, while the last four can be assigned freely. The map in Example 2.2.7 has none. In the isometric case, the node fibers are singletons and the compatible map is precisely the permutation in Lemma 2.1.10.

Example 2.2.10. We now construct a semiconjugacy from a twelve-node network to a four-node network, using four different aggregation rules. Partition [12] into

$$B_1 = \{1,2,3\}, \quad B_2 = \{4,5,6\}, \quad B_3 = \{7,8,9\}, \quad B_4 = \{10,11,12\},$$

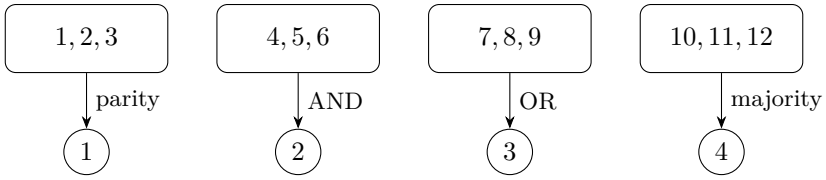
and put

$$\begin{aligned} P(x) &= x_1 + x_2 + x_3, & A(x) &= x_4x_5x_6, \\ O(x) &= x_7 \vee x_8 \vee x_9, & M(x) &= (x_{10}x_{11}) \vee (x_{10}x_{12}) \vee (x_{11}x_{12}). \end{aligned}$$

Thus M is the majority function on the last three inputs. Set

$$h(x) = (P(x), A(x), O(x), M(x)), \qquad \phi(i) = a \quad \text{for } i \in B_a.$$

The assignment of nodes is displayed below; the arrows describe ϕ, rather than wiring edges.



Each rule takes a constant triple (t, t, t) to t, so

$$s(y) = (y_1, y_1, y_1, y_2, y_2, y_2, y_3, y_3, y_3, y_4, y_4, y_4)$$

satisfies $h(s(y)) = y$. Hence h is surjective, and Proposition 2.2.9 shows that it is a contraction with compatible node map ϕ. Every source coordinate is essential in its aggregation rule, so ϕ is unique. This map reduces 4096 states to 16 states. Its state fibers need not have equal sizes: the fibers over $(0, 1, 0, 0)$ and $(0, 0, 1, 0)$ have sizes $4 \cdot 1 \cdot 1 \cdot 4 = 16$ and $4 \cdot 7 \cdot 7 \cdot 4 = 784$, respectively.

Define the target network by

$$g(y) = \big(y_1 y_2 + y_3,\ y_1 \vee y_4,\ y_2 + y_4,\ y_1(y_3 \vee y_4)\big). \tag{2.10}$$

For the construction of f, abbreviate the four prescribed aggregate updates as

$$u_1 = PA + O, \qquad u_2 = P \vee M, \qquad u_3 = A + M, \qquad u_4 = P(O \vee M).$$

These are explicit functions of x, namely the components of $g(h(x))$. Define f block by block:

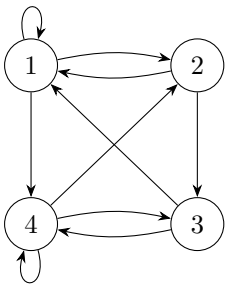
$$\begin{aligned}
(f_1, f_2, f_3) &= (x_{10}, x_{11}, u_1 + x_{10} + x_{11}),\\
(f_4, f_5, f_6) &= (u_2 \vee x_7, u_2 \vee \overline{x}_7, u_2 \vee x_8),\\
(f_7, f_8, f_9) &= (u_3 x_1, u_3 \overline{x}_1, u_3 x_2),\\
(f_{10}, f_{11}, f_{12}) &= (u_4, x_4, \overline{x}_4).
\end{aligned}$$

Applying the four aggregation rules to these updates gives

$$\begin{aligned}
P(f(x)) &= x_{10} + x_{11} + u_1 + x_{10} + x_{11} = u_1,\\
A(f(x)) &= (u_2 \vee x_7)(u_2 \vee \overline{x}_7)(u_2 \vee x_8) = u_2,\\
O(f(x)) &= (u_3 x_1) \vee (u_3 \overline{x}_1) \vee (u_3 x_2) = u_3,\\
M(f(x)) &= \mathrm{maj}(u_4, x_4, \overline{x}_4) = u_4.
\end{aligned}$$

The last equality holds because the last two arguments are opposite, leaving the first argument to determine the majority. Thus $hf = gh$. None of the twelve coordinate functions of f is constant.

All four target nodes interact. The wiring diagram of g and one lift of each of its ten edges are shown below. Each lift is an essential dependence obtained directly from the displayed coordinate functions of f.



$W(g)$

Edge in $W(g)$	A lift in $W(f)$
$1 \to 1$	$1 \to 3$
$2 \to 1$	$4 \to 3$
$3 \to 1$	$7 \to 3$
$1 \to 2$	$1 \to 4$
$4 \to 2$	$10 \to 4$
$2 \to 3$	$4 \to 7$
$4 \to 3$	$10 \to 7$
$1 \to 4$	$1 \to 10$
$3 \to 4$	$7 \to 10$
$4 \to 4$	$10 \to 10$

The reverse implication fails. For instance, $10 \to 1$ is an edge of $W(f)$ because $f_1 = x_{10}$. Its endpoints have images 4 and 1, but $4 \to 1$ is absent from $W(g)$: the occurrences of x_{10} cancel in the first aggregate. Similarly, the source edges $7 \to 4$, $1 \to 7$, and $4 \to 11$ have images $3 \to 2$, $1 \to 3$, and $2 \to 4$, respectively, all absent from $W(g)$. The AND, OR, and majority computations above explain their disappearance. In particular, a compatible node map need not be a graph homomorphism between the wiring diagrams.

The construction is not limited to twelve nodes or to these four rules. Let $[n] = B_1 \sqcup \cdots \sqcup B_m$ be any partition into nonempty blocks, and choose any nonconstant functions $H_a \colon \mathbb{F}_2^{B_a} \to \mathbb{F}_2$. Define h by equation (2.9). Independent choices in the blocks show that h is surjective, and its block assignment is compatible. For each a, choose states $s_a(0), s_a(1)$ with $H_a(s_a(t)) = t$. Given any target network g, define a source network by

$$f(x)|_{B_a} = s_a\big(g_a(h(x))\big), \qquad a \in [m].$$

Then $h_a(f(x)) = g_a(h(x))$ for every a, so h is a contractive semiconjugacy. Both the number of blocks and their sizes can be arbitrarily large. The preceding example allows additional variation within the aggregate fibers, which accounts for its disappearing dependencies.

2.2.4 Lifting edges through node fibers

The examples suggest a one-sided relationship between wiring diagrams: every dependency in the factor should come from at least one dependency between the appropriate source blocks. Several source edges may give the same target edge, and some source dependencies may disappear under aggregation. For a node map ϕ, write

$$(\phi \times \phi)(E(W(f))) = \{(\phi(i), \phi(j)) : (i, j) \in E(W(f))\}.$$

We seek inclusion of $E(W(g))$ in this set. Equality, and hence preservation of every source edge, is not asserted.

Here *edge lifting* means the existence of an edge between the indicated node fibers. A graph fibration additionally requires a graph homomorphism and a unique incoming-edge lift at every prescribed vertex above its target [3, Definition 3.1].

Theorem 2.2.11 (Edge lifting)**.** *Let $f \colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ and $g \colon \mathbb{F}_2^m \to \mathbb{F}_2^m$ be Boolean networks. Suppose that $h \colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ is a contractive semiconjugacy admitting a compatible node map $\phi \colon [n] \to [m]$. Write $B_a = \phi^{-1}(a)$. Then*

$$a \longrightarrow b \text{ in } W(g) \quad \Longrightarrow \quad \begin{array}{c} \text{there exist } i \in B_a \text{ and } j \in B_b \\ \text{such that } i \longrightarrow j \text{ in } W(f). \end{array} \tag{2.11}$$

Equivalently,

$$E(W(g)) \subseteq (\phi \times \phi)(E(W(f))).$$

Self-loops are included: a loop at a lifts to an edge whose two endpoints lie in B_a, and these endpoints need not be equal. Since compatibility already implies contraction, semiconjugacy and compatibility are sufficient hypotheses.

Proof. Suppose $a \to b$ belongs to $W(g)$. By essential dependence, there is $y \in \mathbb{F}_2^m$ such that

$$g_b(y) \neq g_b(y + e_a).$$

Surjectivity gives states $x, z \in \mathbb{F}_2^n$ with $h(x) = y$ and $h(z) = y + e_a$. Define x' by replacing only the coordinates of x in B_a:

$$x'_i = \begin{cases} z_i, & i \in B_a, \\ x_i, & i \notin B_a. \end{cases}$$

Proposition 2.2.9 shows that h_c depends only on B_c. Thus $h_a(x') = h_a(z) = y_a + 1$, while $h_c(x') = h_c(x) = y_c$ for every $c \neq a$. Consequently,

$$h(x') = y + e_a, \qquad x'|_{[n]\setminus B_a} = x|_{[n]\setminus B_a}.$$

Applying the semiconjugacy identity gives

$$h_b(f(x)) = g_b(y) \neq g_b(y + e_a) = h_b(f(x')).$$

Since h_b depends only on B_b, the output states $f(x)$ and $f(x')$ must differ in at least one coordinate $j \in B_b$. Fix such a j.

Join x to x' by changing each coordinate in which they differ exactly once. All these coordinates belong to B_a. Write the resulting path as

$$x = x^{(0)}, x^{(1)}, \ldots, x^{(\ell)} = x'.$$

Since $f_j(x) \neq f_j(x')$, there is a step t for which

$$f_j(x^{(t)}) \neq f_j(x^{(t+1)}).$$

This step has $x^{(t+1)} = x^{(t)} + e_i$ for some $i \in B_a$. Hence f_j depends essentially on x_i, so $i \to j$ in $W(f)$. By construction, $\phi(i) = a$ and $\phi(j) = b$, as required. The argument also applies when $a = b$. □

Chapter 3

Node Maps and Factorization

3.1 Node maps and image factorization

We extend the node-map compatibility condition of the preceding chapter to state maps with proper images. Composition and factorization through these images provide the basis for the quotient theorems and universal properties developed below.

Example 3.1.1. Let $f\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ and $g\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ be the networks

$$f(x_1, x_2, x_3) = (x_3,\, x_1 + x_2,\, x_1 + x_2), \qquad g(y_1, y_2, y_3) = (y_1 + y_2 + y_3,\, y_1 + y_3,\, y_3),$$

and consider the state map

$$h(x_1, x_2, x_3) = (x_1 + x_2,\, x_3,\, 0).$$

Put $u = x_1 + x_2$ and $v = x_3$. Then

$$h(f(x)) = (u + v, u, 0) = g(h(x)).$$

The map h is noninjective, since $h(0, 0, v) = h(1, 1, v)$, and has image

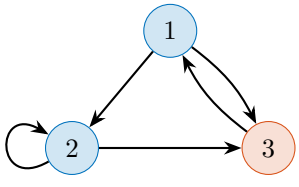
$$S = \{(u, v, 0) : u, v \in \mathbb{F}_2\} \subsetneq \mathbb{F}_2^3.$$

The assignment

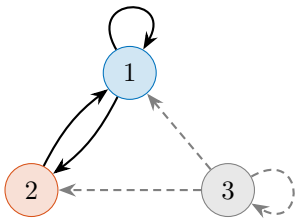
$$\phi(1) = \phi(2) = 1, \qquad \phi(3) = 2$$

satisfies the compatibility condition (2.8).

In the wiring diagrams, colors indicate the node assignment, and gray dashed arrows denote dependencies on y_3, which is zero on S.



$W(f)$



$W(g)$

Edges from target node 3 have no lifts under ϕ; the induced dynamics is determined by $g|_S$.

For arbitrary maps $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ and $\phi\colon [n] \to [m]$, we retain the compatibility condition (2.8) and say that h *admits a node map* when such a ϕ exists. The node fibers $B_a = \phi^{-1}(a)$ may be empty. If h is surjective, every output coordinate is nonconstant, so compatibility forces ϕ to be surjective. A map h may admit several node maps when it ignores some inputs. Throughout this chapter, node counts are positive.

3.1.1 Composition of node maps

Proposition 3.1.2 (Composition of node maps)**.** *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ admit a node map $\phi\colon [n] \to [m]$, and let $k\colon \mathbb{F}_2^m \to \mathbb{F}_2^\ell$ admit a node map $\psi\colon [m] \to [\ell]$. Then $k \circ h$ admits the node map $\psi \circ \phi$.*

Proof. Fix $x \in \mathbb{F}_2^n$ and $i \in [n]$. Compatibility gives either $h(x+e_i) = h(x)$ or $h(x+e_i) = h(x)+e_{\phi(i)}$. In the first case, $k(h(x+e_i)) = k(h(x))$. In the second, compatibility of ψ at the state $h(x)$ gives

$$k(h(x+e_i)) + k(h(x)) \in \{\mathbf{0}, e_{\psi(\phi(i))}\}.$$

This is the required condition for $\psi \circ \phi$. □

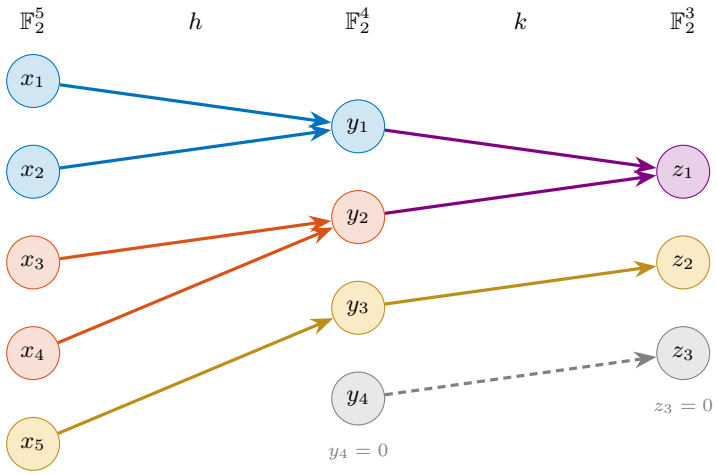
Example 3.1.3. Consider

$$\begin{aligned} h\colon \mathbb{F}_2^5 &\longrightarrow \mathbb{F}_2^4, & h(x_1,x_2,x_3,x_4,x_5) &= (x_1+x_2,\ x_3x_4,\ x_5,\ 0),\\ k\colon \mathbb{F}_2^4 &\longrightarrow \mathbb{F}_2^3, & k(y_1,y_2,y_3,y_4) &= (y_1y_2,\ y_3,\ y_4). \end{aligned}$$

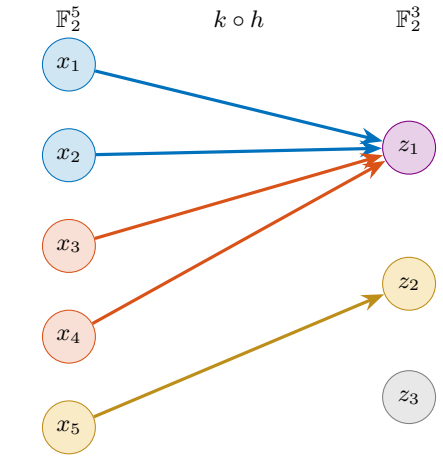
The first map has a node map with fibers $\{1,2\}$, $\{3,4\}$, $\{5\}$, and $\varnothing$. The second has a node map sending $1,2$ to 1, 3 to 2, and 4 to 3. Their composite is

$$(k \circ h)(x) = \big((x_1+x_2)x_3x_4,\ x_5,\ 0\big).$$

The diagram records essential dependencies. The dashed arrow denotes dependence on y_4, which is zero on $\operatorname{im} h$.



The dependence diagram of $k \circ h$ is

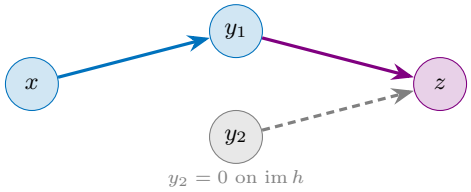


The composite node map has fibers $\{1,2,3,4\}$, $\{5\}$, and $\varnothing$.

Example 3.1.4 (Constant composite of nonconstant maps)**.** Let $h\colon \mathbb{F}_2 \to \mathbb{F}_2^2$ and $k\colon \mathbb{F}_2^2 \to \mathbb{F}_2$ be given by

$$h(x) = (x, 0), \qquad k(y_1, y_2) = y_1 y_2.$$

Both maps admit node maps. Although k depends essentially on both inputs, its restriction to $\operatorname{im} h$ is zero, so $k \circ h = 0$.



3.1.2 Images and essential supports

Proposition 3.1.5 (Image as a coordinate face)**.** *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ admit a node map ϕ. Put*

$$A(h) = \{a \in [m] : h_a \text{ is nonconstant}\}.$$

For $b \notin A(h)$, write c_b for the constant value of h_b. Then

$$\operatorname{im} h = \{y \in \mathbb{F}_2^m : y_b = c_b \text{ for every } b \notin A(h)\}. \tag{3.1}$$

In particular, $|\operatorname{im} h| = 2^{|A(h)|}$.

Proof. Let $B_a = \phi^{-1}(a)$. By condition (2.8), h_a depends only on $x|_{B_a}$. For each $a \in A(h)$, nonconstancy implies that h_a attains both Boolean values. Since the blocks B_a are disjoint, these values can be chosen independently. Thus every assignment to the active coordinates is attained, while the remaining coordinates equal c_b. The image formula and its cardinality follow. □

Thus a map admitting a node map is surjective exactly when all its output coordinates are nonconstant. For a coordinate set $A \subseteq [m]$, we use its increasing order to identify $\mathbb{F}_2^A$ with $\mathbb{F}_2^{|A|}$ when specifying node maps.

For each $a \in [m]$, define the *essential support* of h_a by

$$S_a(h) = \{i \in [n] : h_a \text{ depends essentially on } x_i\}.$$

Let $S = \operatorname{im} h$ and identify S with $\mathbb{F}_2^{A(h)}$ by its active coordinates. For $k\colon \mathbb{F}_2^m \to \mathbb{F}_2^\ell$, write $k_c|_S$ for the resulting Boolean function on $\mathbb{F}_2^{A(h)}$.

Proposition 3.1.6 (Essential supports under composition)**.** *If h admits a node map, then for every output coordinate c of k,*

$$\{i : (k \circ h)_c \text{ depends essentially on } x_i\} = \bigcup_{\substack{a \in A(h) \\ k_c|_S \text{ depends essentially on } a}} S_a(h). \tag{3.2}$$

The map k need not itself admit a node map for this identity.

Proof. Choose a node map ϕ for h and put $B_a = \phi^{-1}(a)$. An input i can change at most one coordinate of h, namely $a = \phi(i)$. If $a \notin A(h)$ or $i \notin S_a(h)$, then it cannot change the composite. If $k_c|_S$ is independent of a, a change in h_a also cannot change the composite. This gives inclusion from left to right.

Conversely, suppose $i \in S_a(h)$ and $k_c|_S$ depends essentially on a. There exist $x \in \mathbb{F}_2^n$ and $y \in S$ such that

$$h_a(x + e_i) \neq h_a(x), \qquad k_c(y + e_a) \neq k_c(y).$$

Since $a \in A(h)$ and S is a coordinate face, $y + e_a \in S$. Replacing y by $y + e_a$ if necessary, we may assume $y_a = h_a(x)$. Choose $w \in \mathbb{F}_2^n$ with $h(w) = y$, and define $z \in \mathbb{F}_2^n$ by

$$z|_{B_a} = x|_{B_a}, \qquad z|_{[n]\setminus B_a} = w|_{[n]\setminus B_a}.$$

Each h_b depends only on the inputs in B_b, and these blocks are disjoint. Thus $h(z) = y$. Since $i \in B_a$, the first inequality also gives $h(z + e_i) = y + e_a$. Consequently,

$$(k \circ h)_c(z + e_i) = k_c(y + e_a) \neq k_c(y) = (k \circ h)_c(z).$$

Hence i is essential in $(k \circ h)_c$, proving the reverse inclusion. □

3.1.3 Factorization through an invariant face

Lemma 3.1.7 (Injective maps with node maps)**.** *An injective Boolean map $j\colon \mathbb{F}_2^r \to \mathbb{F}_2^m$ admits a node map if and only if there is an injection $\iota\colon [r] \to [m]$ such that*

$$j_{\iota(i)}(x) = x_i + \epsilon_i \quad (i \in [r]), \qquad j_b(x) = c_b \quad (b \notin \iota([r])),$$

for constants $\epsilon_i, c_b \in \mathbb{F}_2$. In particular, j preserves Hamming distance and embeds its domain as a coordinate face of dimension r.

Proof. Let $\iota\colon [r] \to [m]$ be a node map for j, and put $B_a = \iota^{-1}(a)$. Fix $a \in \iota([r])$ and $i \in B_a$. For every $x \in \mathbb{F}_2^r$, injectivity gives $j(x + e_i) \neq j(x)$. Compatibility allows only coordinate a to change, so

$$j_a(x + e_i) \neq j_a(x).$$

Thus j_a depends essentially on x_i for every $i \in B_a$.

Now fix all inputs outside B_a and vary the inputs in B_a. Since each j_b depends only on the inputs in B_b, every output coordinate other than a remains constant. The $2^{|B_a|}$ distinct input states therefore have at most two output states. Injectivity requires $2^{|B_a|} \leq 2$. Since B_a is nonempty, $B_a = \{i\}$ for some $i \in [r]$. Hence ι is injective.

Each $j_{\iota(i)}$ depends only on x_i and is nonconstant, so it is x_i or $x_i + 1$. Every output coordinate outside $\iota([r])$ is constant. This proves the stated form, which preserves Hamming distance. Conversely, every map of this form is injective and satisfies condition (2.8) with node map ι. □

Theorem 3.1.8. *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be a nonconstant Boolean map admitting a node map. Put $A = A(h)$ and $S = \operatorname{im} h$. Then h has a factorization, unique up to permutation and complementation of the intermediate coordinates,*

$$\mathbb{F}_2^n \overset{p}{\twoheadrightarrow} \mathbb{F}_2^A \overset{j}{\hookrightarrow} \mathbb{F}_2^m, \qquad h = j \circ p, \tag{3.3}$$

where p and j are respectively surjective and injective, and both admit node maps. Every such factorization has $|A|$ intermediate coordinates. The map j is an isometric embedding with image S.

If $h \circ f = g \circ h$ for Boolean networks f and g, then S is invariant under g. Write $j^{-1}\colon S \to \mathbb{F}_2^A$ for the inverse of j onto its image. The induced network $\bar{g} = j^{-1} \circ g \circ j$ on $\mathbb{F}_2^A$ satisfies

$$p \circ f = \bar{g} \circ p, \qquad j \circ \bar{g} = g \circ j. \tag{3.4}$$

Proof. Define $p(x) = (h_a(x))_{a \in A}$. Proposition 3.1.5 shows that the active coordinates of h take every value in $\mathbb{F}_2^A$ independently, so p is surjective. The original node map for h assigns every essential input of h_a to a. Inputs ignored by all h_a can be assigned to any node in A, which is nonempty because h is nonconstant. This gives a node map for p. Define $j(u)$ by setting $j(u)_a = u_a$ for $a \in A$ and $j(u)_b = c_b$ for $b \notin A$, where c_b is the constant value of h_b. It has the inclusion $A \hookrightarrow [m]$ as a node map and preserves Hamming distance. By construction $j \circ p = h$ and $j(\mathbb{F}_2^A) = S$.

If $y = h(x) \in S$, then $g(y) = g(h(x)) = h(f(x)) \in S$, so S is invariant. Consequently $\bar{g}$ is well defined. From $jpf = hf = gh = gjp = j\bar{g}p$ and injectivity of j, we obtain $pf = \bar{g}p$. The second identity in (3.4) follows from the definition of $\bar{g}$.

For uniqueness, suppose $h = j' \circ p'$ with $p'\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ surjective and $j'\colon \mathbb{F}_2^r \to \mathbb{F}_2^m$ injective, both admitting node maps. Surjectivity gives $\operatorname{im} j' = \operatorname{im} h = S$. By Lemma 3.1.7, j' maps onto a face of dimension r. Since S has dimension $|A|$, we have $r = |A|$. The map $u = j^{-1} \circ j'$ is an isometric bijection of the intermediate cubes, hence a coordinate permutation followed by coordinate complementations by Theorem 2.1.14. Thus $j' = j \circ u$ and $p' = u^{-1} \circ p$. □

For Example 3.1.1, the factorization is explicit:

$$p(x_1, x_2, x_3) = (x_1 + x_2, x_3), \qquad j(u, v) = (u, v, 0), \qquad \bar{g}(u, v) = (u + v, u).$$

The first map aggregates source nodes 1 and 2; the second places the resulting two-node dynamics on the invariant face $y_3 = 0$.

Remark 3.1.9. If h is constant, then $h \circ f = g \circ h$ holds exactly when its value is a fixed point of g. Its image is a zero-dimensional face, but there is no node map $[n] \to [0]$ for $n > 0$. Thus the factorization into maps admitting node maps in Theorem 3.1.8 requires h to be nonconstant.

The factorization is determined by h and may change the assignment of ignored inputs. For example, $h(x_1, x_2) = (x_1, 0)$ admits the node map $\phi(1) = 1$, $\phi(2) = 2$. Factoring through $p(x_1, x_2) = x_1$ instead yields a composite node map assigning both inputs to target node 1.

3.1.4 Quotients and homomorphism theorems

For a state map $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$, define its *fiber relation* by

$$E_h = \{(x, x') \in \mathbb{F}_2^n \times \mathbb{F}_2^n : h(x) = h(x')\}.$$

Its equivalence classes are the fibers of h. If $h \circ f = g \circ h$, then E_h is forward invariant, so $[x] \mapsto [f(x)]$ defines the quotient dynamics. The following descent theorem establishes compatibility with node maps.

Theorem 3.1.10 (Descent through a compatible quotient)**.** *Let $p\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be surjective and admit a node map, and let $k\colon \mathbb{F}_2^n \to \mathbb{F}_2^\ell$ admit a node map. Then*

$$E_p \subseteq E_k \quad \Longleftrightarrow \quad \text{there is a unique map } u\colon \mathbb{F}_2^m \to \mathbb{F}_2^\ell \text{ with } k = u \circ p.$$

Whenever it exists, u admits a node map. It is surjective if k is surjective. If, in addition, $p\circ f = g\circ p$ and $k \circ f = r \circ k$, then $u \circ g = r \circ u$.

Proof. If $k = u \circ p$, then $E_p \subseteq E_k$. Conversely, assume this inclusion and define $u(p(x)) = k(x)$. The inclusion makes u well defined, and surjectivity of p gives uniqueness. Also, $u(\mathbb{F}_2^m) = k(\mathbb{F}_2^n)$, proving the surjectivity assertion.

Since p is surjective, its essential supports $S_a(p)$ are nonempty; compatibility makes them pairwise disjoint. Applying Proposition 3.1.6 to $k = u \circ p$ gives, for each $c \in [\ell]$,

$$S_c(k) = \bigcup_{a \in S_c(u)} S_a(p).$$

If some input coordinate a of u were essential in two different outputs u_c and u_d, then every input in the nonempty set $S_a(p)$ would be essential in both k_c and k_d. This contradicts the existence of a node map for k. Thus each input of u is essential in at most one output. Assign that input to its unique affected output, or assign it arbitrarily if no output depends on it. This assignment satisfies condition (2.8), so u admits a node map.

Finally, $u \circ g \circ p = u \circ p \circ f = k \circ f = r \circ k = r \circ u \circ p$. Surjectivity of p gives $u \circ g = r \circ u$. □

The image face supplies Boolean coordinates for the quotient set.

Theorem 3.1.11 (Fundamental homomorphism theorem for node maps)**.** *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be nonconstant, admit a node map, and satisfy $h \circ f = g \circ h$. Put $S = \operatorname{im} h$. The map*

$$\Theta\colon \mathbb{F}_2^n/E_h \longrightarrow S, \qquad [x] \longmapsto h(x),$$

is a bijection intertwining the quotient dynamics $[x] \mapsto [f(x)]$ with $g|_S$. Moreover, S is an invariant coordinate face, and Theorem 3.1.8 identifies this quotient with a Boolean network on $\mathbb{F}_2^{A(h)}$ through a surjective map admitting a node map. In particular, when h is surjective, the quotient is the whole target network g.

Proof. The definition of E_h makes Θ well defined and injective; its codomain is the image, so it is surjective. If $h(x) = h(x')$, then $h(f(x)) = g(h(x)) = g(h(x')) = h(f(x'))$, so quotient dynamics is well defined. The same identity gives

$$\Theta([f(x)]) = h(f(x)) = g(h(x)) = g(\Theta([x])).$$

Proposition 3.1.5 and Theorem 3.1.8 give the claimed face and its Boolean coordinates. □

Example 3.1.12 (A nonlinear quotient with unequal fibers)**.** Consider

$$p(x_1, x_2, x_3) = (x_1x_2, x_3), \qquad f(x_1, x_2, x_3) = (x_1, x_2, x_3 + 1), \qquad g(u, v) = (u, v + 1).$$

The map p is surjective, has node fibers $\{1, 2\}$ and $\{3\}$, and satisfies $p \circ f = g \circ p$. For each $v \in \mathbb{F}_2$, the fibers over $(0, v)$ and $(1, v)$ have sizes three and one, respectively. Theorem 3.1.11 identifies the quotient with g, which has two 2-cycles.

If surjective maps $p\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ and $q\colon \mathbb{F}_2^n \to \mathbb{F}_2^\ell$ admit node maps and satisfy $E_p = E_q$, Theorem 3.1.10 gives $q = u \circ p$ and $p = v \circ q$. Surjectivity makes u and v inverse maps. By Lemma 3.1.7, $m = \ell$ and u is a coordinate permutation followed by coordinate complementations. Thus the fiber relation determines the quotient coordinates up to an isometry.

Theorem 3.1.13 (Nested-quotient theorem)**.** *Let $p\colon (\mathbb{F}_2^n, f) \to (\mathbb{F}_2^m, g)$ and $q\colon (\mathbb{F}_2^n, f) \to (\mathbb{F}_2^\ell, r)$ be surjective maps admitting node maps and satisfying the semiconjugacy identities. If $E_p \subseteq E_q$, there is a unique surjective map $u\colon (\mathbb{F}_2^m, g) \to (\mathbb{F}_2^\ell, r)$ admitting a node map such that $q = u \circ p$. The relation on $\mathbb{F}_2^m$ given by*

$$y \; E_q/E_p \; y' \quad \Longleftrightarrow \quad q(x) = q(x') \quad \text{for any } x \in p^{-1}(y),\ x' \in p^{-1}(y')$$

is well defined and equals E_u. Using p to identify $\mathbb{F}_2^n/E_p$ with $\mathbb{F}_2^m$, we obtain, with their induced dynamics,

$$\big(\mathbb{F}_2^n/E_p\big)/(E_q/E_p) \;\cong\; \mathbb{F}_2^n/E_q, \qquad [[x]] \longmapsto [x].$$

Both sides acquire the same ℓ-node quotient network through their maps to $\mathbb{F}_2^\ell$.

Proof. Theorem 3.1.10 gives a unique surjective map u admitting a node map with $q = u \circ p$ and $u \circ g = r \circ u$. Thus the displayed relation is $u(y) = u(y')$, independent of the choices of x, x'. The map on quotient classes is well defined because $u(p(x)) = q(x)$ and bijective because both quotients identify exactly the pairs in E_q. The intertwining identities give the same induced update on both quotients, with Boolean coordinates supplied by u and q. □

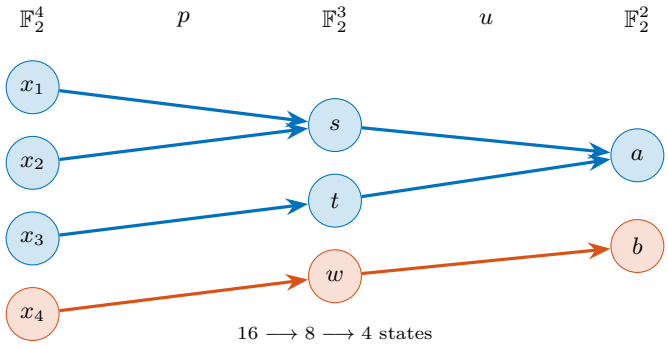
Example 3.1.14 (Successive quotients)**.** Take $p\colon \mathbb{F}_2^4 \to \mathbb{F}_2^3$ and $u\colon \mathbb{F}_2^3 \to \mathbb{F}_2^2$ to be

$$p(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_3, x_4) = (s, t, w), \qquad u(s, t, w) = (s + t, w) = (a, b).$$

The source, intermediate, and final networks are

$$f(x) = (x_1 + x_4,\ x_1,\ x_1 + x_2 + x_3,\ x_1 + x_2 + x_3),$$
$$g(s, t, w) = (w,\ s + t,\ s + t), \qquad r(a, b) = (a + b, a).$$

Direct substitution gives $p \circ f = g \circ p$ and $u \circ g = r \circ u$. The diagram records the node assignments for both maps.



The composite $(u \circ p)(x) = (x_1 + x_2 + x_3, x_4)$ induces the quotient update r, with fixed point 00 and cycle $01 \to 10 \to 11 \to 01$.

An invariant coordinate face of positive dimension carries a Boolean network on its free coordinates.

Proposition 3.1.15 (Restriction to an invariant face)**.** *Let $p\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be a surjective map admitting a node map and satisfying $p \circ f = g \circ p$. If $T \subseteq \mathbb{F}_2^n$ is an invariant coordinate face, then $p(T)$ is an invariant coordinate face of $\mathbb{F}_2^m$. The restriction $p|_T$ gives a bijection of dynamical systems*

$$T/(E_p|_{T\times T}) \;\cong\; p(T), \qquad [x] \longmapsto p(x).$$

If $p(T)$ has positive dimension, the restriction $p|_T$ is a semiconjugacy admitting a node map when T and $p(T)$ are expressed in their free coordinates.

Proof. If T is a singleton, then $p(T)$ is a singleton and the assertions are immediate. Otherwise, T has positive dimension. The inclusion of T into $\mathbb{F}_2^n$, expressed in its free coordinates, is an injective map admitting a node map by Lemma 3.1.7. Its composition with p admits a node map by Proposition 3.1.2; hence its image $p(T)$ is a coordinate face by Proposition 3.1.5. For $y = p(x)$ with $x \in T$, we have $g(y) = p(f(x)) \in p(T)$ because T is invariant. The displayed bijection and its intertwining identity follow as in Theorem 3.1.11. When $p(T)$ has positive dimension, factor the restricted map through its active coordinates using Theorem 3.1.8. □

Example 3.1.16 (Saturation of an invariant face)**.** In Example 3.1.12, the invariant face $T = \{x_1 = 0\}$ has image $p(T) = \{u = 0\}$. In free coordinates, $p|_T$ is the projection $(x_2, x_3) \mapsto x_3$, with quotient update $v \mapsto v + 1$.

The saturation of T is

$$p^{-1}(p(T)) = \{(x_1, x_2, x_3) : x_1 x_2 = 0\}.$$

This set is invariant under f and has six states, so it is not a coordinate face. Hence coordinate faces are not closed under saturation.

3.2 Universal properties and tensor products

The image factorization has universal properties determined by fiber inclusion and image containment. We establish these properties, examine their compatibility with prescribed updates, and study their behavior under tensor products.

3.2.1 Universal properties of the image

Theorem 3.2.1 (Universal properties of the image factorization)**.** *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ be nonconstant and admit a node map. Write*

$$h = j \circ p, \qquad p\colon \mathbb{F}_2^n \to \mathbb{F}_2^A, \qquad j\colon \mathbb{F}_2^A \to \mathbb{F}_2^m, \qquad A = A(h),$$

for the factorization in Theorem 3.1.8. Then:

(i) If $k\colon \mathbb{F}_2^n \to \mathbb{F}_2^\ell$ admits a node map, there is a unique map $u\colon \mathbb{F}_2^A \to \mathbb{F}_2^\ell$ with $k = u \circ p$ if and only if $E_h \subseteq E_k$. Whenever it exists, u admits a node map.

(ii) If $k\colon \mathbb{F}_2^\ell \to \mathbb{F}_2^m$ admits a node map, there is a unique map $v\colon \mathbb{F}_2^\ell \to \mathbb{F}_2^A$ with $k = j \circ v$ if and only if $\operatorname{im} k \subseteq \operatorname{im} h$. Whenever it exists, v admits a node map.

If $h \circ f = g \circ h$, let $\bar{g}$ be the induced update on $\mathbb{F}_2^A$. In (i), $k \circ f = r \circ k$ implies $u \circ \bar{g} = r \circ u$. In (ii), $k \circ r = g \circ k$ implies $v \circ r = \bar{g} \circ v$.

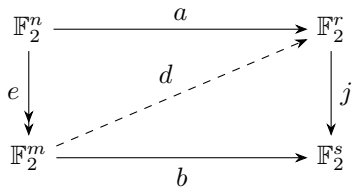
Proof. Since j is injective, $E_h = E_p$. Part (i), including the intertwining identity, follows from Theorem 3.1.10.

For (ii), the equation $k = j \circ v$ requires $\operatorname{im} k \subseteq \operatorname{im} j = \operatorname{im} h$. Conversely, assume this inclusion. Let $\pi_A \colon \mathbb{F}_2^m \to \mathbb{F}_2^A$ be the coordinate projection, and put $v = \pi_A \circ k$. Since A is nonempty, π_A admits a node map, as does v by composition. The image inclusion gives $j \circ v = k$, and injectivity of j proves uniqueness. Finally,

$$j \circ v \circ r = k \circ r = g \circ k = g \circ j \circ v = j \circ \bar{g} \circ v.$$

Cancelling j gives the last assertion. □

Theorem 3.2.2 (Lifting through a quotient)**.** *Suppose all four maps in the following square admit node maps and satisfy $j \circ a = b \circ e$, with e surjective and j injective:*



There is a unique map $d \colon \mathbb{F}_2^m \to \mathbb{F}_2^r$ with

$$d \circ e = a, \qquad j \circ d = b,$$

and this map admits a node map. If updates f, g, u on $\mathbb{F}_2^n, \mathbb{F}_2^m, \mathbb{F}_2^r$, respectively, satisfy $e \circ f = g \circ e$ and $a \circ f = u \circ a$, then $d \circ g = u \circ d$.

Proof. If $e(x) = e(x')$, commutativity gives $j(a(x)) = j(a(x'))$. Injectivity of j gives $a(x) = a(x')$, so $E_e \subseteq E_a$. Theorem 3.1.10 supplies a unique map d admitting a node map with $d \circ e = a$, together with the intertwining identity. Finally,

$$j \circ d \circ e = j \circ a = b \circ e.$$

Surjectivity of e gives $j \circ d = b$. □

Theorem 3.2.3 (Injectivity and unique lifting)**.** *A Boolean map $j \colon \mathbb{F}_2^r \to \mathbb{F}_2^s$ admitting a node map is injective if and only if every square of the form in Theorem 3.2.2, with this j and a surjective e, has a unique map d admitting a node map and satisfying the two lifting identities. The equivalence also holds for a map j intertwining prescribed updates, with all maps in the square and the lift required to intertwine the corresponding updates.*

Proof. The forward implication is Theorem 3.2.2. Conversely, first suppose j is nonconstant and write $j = i \circ p$ as in Theorem 3.1.8, with p surjective and i injective. Apply the assumed lifting property to

$$j \circ \mathrm{Id}_{\mathbb{F}_2^r} = i \circ p.$$

It gives $d \circ p = \mathrm{Id}_{\mathbb{F}_2^r}$. Thus p is injective, and so is $j = i \circ p$.

If j is constant, use the two projections

$$e(x, y) = x, \qquad a(x, y) = y \qquad \text{from } \mathbb{F}_2^{2r} \text{ to } \mathbb{F}_2^r,$$

and put $b = j$. Then $j \circ a = b \circ e$, but $d \circ e = a$ would require $d(x) = y$ for all x, y, which is impossible. Both projections admit node maps, so this excludes the constant case.

For prescribed updates, the nonconstant argument uses the induced update on the image face. In the constant case, if f is the source update of j, give $\mathbb{F}_2^{2r}$ the update $(x, y) \mapsto (f(x), f(y))$. Both projections intertwine the updates, and the same contradiction applies. □

3.2.2 One-sided inverses and retractions

Proposition 3.2.4. *If $p\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ is surjective and admits a node map, there is a map $s\colon \mathbb{F}_2^r \to \mathbb{F}_2^n$ admitting a node map such that $p \circ s = \mathrm{Id}_{\mathbb{F}_2^r}$. The map s is an isometric embedding onto a coordinate face. If $j\colon \mathbb{F}_2^r \to \mathbb{F}_2^m$ is injective and admits a node map, there is a map $q\colon \mathbb{F}_2^m \to \mathbb{F}_2^r$ admitting a node map such that $q \circ j = \mathrm{Id}_{\mathbb{F}_2^r}$.*

Proof. Choose a node map for p, with blocks $B_1, \ldots, B_r$. Each p_a is nonconstant on its block. A path joining an input with value zero to one with value one therefore contains an edge on which p_a changes. Choose such an edge in each block, and let $i_a \in B_a$ be its changing coordinate. Fix the other coordinates of that block at their values on the chosen edge. On this edge, p_a is either x_{i_a} or $x_{i_a} + 1$.

Given $y \in \mathbb{F}_2^r$, choose the endpoint in block B_a for which $p_a = y_a$. The disjoint blocks make these choices independent and define $s(y)$. Its only changing coordinates are the distinct i_a, each equal to y_a or $y_a + 1$. Thus s admits a node map, preserves Hamming distance, and satisfies $p \circ s = \mathrm{Id}$.

For the second assertion, use Lemma 3.1.7. If $j_{\iota(a)}(x) = x_a + \epsilon_a$, put $q_a(y) = y_{\iota(a)} + \epsilon_a$. These components use distinct input coordinates, so q admits a node map, and $q \circ j = \mathrm{Id}$. □

These one-sided inverses need not intertwine network updates. For example, let

$$f(x_1, x_2) = (x_2, 1 + x_1), \qquad p(x_1, x_2) = x_1 + x_2, \qquad g(t) = t + 1.$$

Then $p \circ f = g \circ p$. A map s satisfying both $p \circ s = \mathrm{Id}$ and $s \circ g = f \circ s$ would have an invariant two-state image under f. This is impossible because f has the single cycle $00 \to 01 \to 11 \to 10 \to 00$.

Proposition 3.2.5 (Retraction onto the image)**.** *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^n$ admit a node map and satisfy $h \circ h = h$. If h is nonconstant, its image factorization $h = j \circ p$ satisfies*

$$p \circ j = \mathrm{Id}_{\mathbb{F}_2^{A(h)}}.$$

If $h \circ f = f \circ h$, both factors intertwine f with the induced update on the image. If h is constant, there are no maps through a positive-dimensional intermediate cube satisfying both $h = j \circ p$ and $p \circ j = \mathrm{Id}$.

Proof. For nonconstant h, substitute $h = j \circ p$ into $h \circ h = h$ to obtain

$$j \circ p \circ j \circ p = j \circ p.$$

Cancel the injective j and then the surjective p to obtain $p \circ j = \mathrm{Id}$. The assertion about updates follows from Theorem 3.1.8. In any factorization with $p \circ j = \mathrm{Id}$, the map p is surjective and j is injective, so $\operatorname{im} h = \operatorname{im} j$. If h is constant, the intermediate cube must therefore have a single state. □

3.2.3 Cancellation and prescribed dynamics

For maps intertwining prescribed updates, surjectivity is equivalent to right cancellation.

Proposition 3.2.6 (Right cancellation)**.** *Let $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ admit a node map and satisfy $h\circ f = g\circ h$. Then h is surjective if and only if the following property holds: for every Boolean update $r\colon \mathbb{F}_2^s \to \mathbb{F}_2^s$ and all maps $k, \ell\colon \mathbb{F}_2^m \to \mathbb{F}_2^s$ admitting node maps with*

$$k \circ g = r \circ k, \qquad \ell \circ g = r \circ \ell,$$

the equality $k \circ h = \ell \circ h$ implies $k = \ell$.

Proof. Surjectivity gives the stated cancellation. Conversely, suppose $S = \operatorname{im} h$ is proper. The identity $h \circ f = g \circ h$ makes S invariant under g.

If there is $z \notin S$ with no preimage under g, let H be the indicator of $\{z\}$. Then $H \circ g = 0$. The maps H and 0 are distinct scalar maps, both intertwining g with the constant-zero update, and their composites with h agree.

Otherwise every state outside S has a preimage under g, and none of these preimages lies in S. Since the complement is finite, g permutes it. Its indicator H satisfies $H \circ g = H$. Thus H and 0 are distinct scalar maps intertwining g with the identity update, again agreeing after composition with h. Every scalar Boolean map admits a node map, so both cases contradict the stated property. □

Injectivity implies left cancellation. Among Boolean maps admitting node maps, the converse follows by considering constant maps: if $h(x) = h(x')$, the constant maps from $\mathbb{F}_2$ with values x and x' have equal composites with h. For prescribed updates, the converse can fail, as the following lemma and example establish.

Lemma 3.2.7 (Rigidity of majority aggregation)**.** *Let $a, b\colon \mathbb{F}_2^n \to \mathbb{F}_2^3$ be surjective maps admitting node maps. If*

$$\operatorname{maj}(a_1, a_2, a_3) = \operatorname{maj}(b_1, b_2, b_3),$$

where $\operatorname{maj}(u_1, u_2, u_3) = u_1u_2 + u_1u_3 + u_2u_3$*, then $b = P_\sigma \circ a$ for a permutation σ of* $[3]$.

Proof. Each triple consists of nonconstant functions with pairwise disjoint essential supports. By Proposition 3.1.6, the essential support of their common majority function is the union of the supports in either triple. Let its size be N, so $N \geq 3$. We may omit the ignored inputs and work on these N variables.

The three-input majority function is prime for disjoint-support decomposition. Indeed, replacing any pair of its inputs by a single Boolean value would need to distinguish the three residual functions $0, t, 1$, obtained when the pair has values $00, 01, 11$. A single Boolean value cannot distinguish all three.

If $N > 3$, uniqueness of disjoint-support decompositions with a prime outer function on more than two inputs implies that the two triples agree up to permutation and complementation [1, Section 2.1]. If $N = 3$, every component in either triple is a literal on one of the same three input variables, and this conclusion holds directly. Consequently,

$$b = P_\sigma \circ a + \epsilon$$

for a permutation σ of $[3]$ and a vector $\epsilon \in \mathbb{F}_2^3$. Surjectivity of a and symmetry of majority give

$$\operatorname{maj}(u + \epsilon) = \operatorname{maj}(u) \qquad \text{for all } u \in \mathbb{F}_2^3.$$

Taking $u = \mathbf{0}$ shows that ϵ has at most one nonzero coordinate. If $\epsilon = e_j$, take $u = e_k$ with $k \neq j$. Then $\operatorname{maj}(u) = 0$ but $\operatorname{maj}(u + \epsilon) = 1$, a contradiction. Hence $\epsilon = \mathbf{0}$. □

Example 3.2.8 (Left cancellation without injectivity)**.** Define $f\colon \mathbb{F}_2^3 \to \mathbb{F}_2^3$ by the single cycle

$$000 \to 111 \to 001 \to 110 \to 010 \to 101 \to 100 \to 011 \to 000,$$

and put $g(t) = t+1$ and $p(x) = \mathrm{maj}(x_1, x_2, x_3)$. Majority alternates along the cycle, so $p \circ f = g \circ p$. The map p admits the node map sending all three nodes to the sole target node. It is surjective and has two state fibers of size four.

Let $r\colon \mathbb{F}_2^s \to \mathbb{F}_2^s$ be any update, and suppose $a, b\colon \mathbb{F}_2^s \to \mathbb{F}_2^3$ admit node maps and satisfy

$$a \circ r = f \circ a, \qquad b \circ r = f \circ b, \qquad p \circ a = p \circ b.$$

Their images are nonempty invariant faces. A single cycle through all eight states has no proper nonempty invariant subset, so a and b are surjective. Lemma 3.2.7 gives $b = P_\sigma \circ a$. The intertwining identities imply

$$P_\sigma \circ f \circ a = P_\sigma \circ a \circ r = b \circ r = f \circ b = f \circ P_\sigma \circ a.$$

Surjectivity of a gives $P_\sigma \circ f = f \circ P_\sigma$. Every coordinate permutation fixes 000. Commutation with f forces P_σ to fix the whole cycle, so $P_\sigma = \mathrm{Id}$ and $a = b$.

Thus p satisfies left cancellation among maps intertwining the prescribed updates, although it is noninjective. Without the intertwining requirement, the constant maps with values 000 and 001 violate cancellation.

3.2.4 Tensor products of maps and networks

Definition 3.2.9. For Boolean maps $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ and $k\colon \mathbb{F}_2^m \to \mathbb{F}_2^s$, define

$$h \otimes k\colon \mathbb{F}_2^{n+m} \to \mathbb{F}_2^{r+s}, \qquad (h \otimes k)(x, y) = (h(x), k(y)).$$

For networks $X = (\mathbb{F}_2^n, f)$ and $Y = (\mathbb{F}_2^m, g)$, write

$$X \otimes Y = (\mathbb{F}_2^{n+m}, f \otimes g).$$

The coordinates of the first factor precede those of the second.

Proposition 3.2.10. *If h and k admit node maps, so does $h \otimes k$. This tensor is injective exactly when both factors are injective, and surjective exactly when both are surjective. For maps of the appropriate dimensions,*

$$(h' \otimes k') \circ (h \otimes k) = (h' \circ h) \otimes (k' \circ k).$$

In particular, if $h \circ f = f' \circ h$ and $k \circ g = g' \circ k$, then

$$(h \otimes k) \circ (f \otimes g) = (f' \otimes g') \circ (h \otimes k).$$

Regrouping or exchanging coordinate blocks gives Hamming isometric conjugacies between the corresponding tensor networks.

Proof. If $\phi\colon [n] \to [r]$ and $\psi\colon [m] \to [s]$ are node maps for h and k, respectively, a node map for their tensor is

$$i \longmapsto \begin{cases} \phi(i), & 1 \le i \le n, \\ r + \psi(i-n), & n < i \le n+m. \end{cases}$$

Injectivity and surjectivity follow by considering each output block separately. The composition identity follows by evaluation at (x, y) and implies the intertwining identity. Regrouping (x, y, z) and exchanging (x, y) with (y, x) are coordinate permutations that intertwine the corresponding updates. □

Theorem 3.2.11 (Tensor products and image factorization)**.** *Let h and k be nonconstant Boolean maps admitting node maps, with image factorizations $h = j_h \circ p_h$ and $k = j_k \circ p_k$. Then*

$$h \otimes k = (j_h \otimes j_k) \circ (p_h \otimes p_k)$$

is a factorization through the image of $h \otimes k$, with a surjective first map and an injective second map, both admitting node maps. The intermediate cube has $|A(h)| + |A(k)|$ coordinates. If the maps intertwine network updates, the intermediate update is the tensor of the two induced updates.

For surjective maps $p\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ and $q\colon \mathbb{F}_2^m \to \mathbb{F}_2^s$ admitting node maps, the fiber relations satisfy

$$(x, y)\ E_{p\otimes q}\ (x', y') \quad \Longleftrightarrow \quad x\ E_p\ x' \text{ and } y\ E_q\ y'.$$

If $p \circ f = \bar{f} \circ p$ and $q \circ g = \bar{g} \circ q$, the resulting quotient update is $\bar{f} \otimes \bar{g}$.

Proof. Proposition 3.2.10 gives the displayed factorization and the properties of its two maps. Moreover,

$$\operatorname{im}(h \otimes k) = \operatorname{im} h \times \operatorname{im} k.$$

This face has the stated dimension, and its induced update is the tensor of the two induced updates. Theorem 3.1.8 identifies the intermediate coordinates up to permutation and complementation. The fiber identity follows from equality in each output block, and the quotient update follows from Proposition 3.2.10. □

The induced bijection of quotient sets is

$$\mathbb{F}_2^{n+m}/E_{p\otimes q} \longrightarrow (\mathbb{F}_2^n/E_p) \times (\mathbb{F}_2^m/E_q), \qquad [(x, y)] \longmapsto ([x], [y]).$$

It intertwines the quotient updates whenever the original maps intertwine their updates.

Example 3.2.12. Consider the quotient $p(x_1, x_2) = x_1 + x_2$ from

$$f(x_1, x_2) = (x_2, 1 + x_1)$$

to the one-node flip $\bar{f}(u) = u + 1$, and the quotient $q(y_1, y_2) = y_1 y_2$ from the two-node identity network to the one-node identity network. Then

$$(p \otimes q)(x_1, x_2, y_1, y_2) = (x_1 + x_2, y_1 y_2)$$

reduces the four-node update

$$(x_1, x_2, y_1, y_2) \longmapsto (x_2, 1 + x_1, y_1, y_2)$$

to $(u, v) \mapsto (u + 1, v)$. Its fibers have size six when $v = 0$ and size two when $v = 1$. The quotient has two 2-cycles.

Write $I_n = (\mathbb{F}_2^n, \mathrm{Id}_{\mathbb{F}_2^n})$ for the n-node identity network. A *passive node* is a factor I_1. Tensoring with $\mathrm{Id}_{\mathbb{F}_2}$ gives a nonconstant map, so the image factorization applies even when the original map is constant.

Theorem 3.2.13 (Factorization with a passive node)**.** *For every Boolean map $h\colon \mathbb{F}_2^n \to \mathbb{F}_2^m$ admitting a node map,*

$$h \otimes \mathrm{Id}_{\mathbb{F}_2}\colon \mathbb{F}_2^{n+1} \to \mathbb{F}_2^{m+1}$$

factors through a cube with $|A(h)| + 1$ coordinates as a surjective map followed by an injective map, both admitting node maps. Its image is $\operatorname{im} h \times \mathbb{F}_2$. If $h \circ f = g \circ h$, the factors intertwine the updates $f \otimes \mathrm{Id}_{\mathbb{F}_2}$ and $g \otimes \mathrm{Id}_{\mathbb{F}_2}$ with the induced intermediate update.

Proof. Since $h \otimes \mathrm{Id}_{\mathbb{F}_2}$ is nonconstant, Theorem 3.1.8 applies. Proposition 3.1.5 gives the image and dimension.

For nonconstant $h = j \circ p$, the factors are $p \otimes \mathrm{Id}_{\mathbb{F}_2}$ and $j \otimes \mathrm{Id}_{\mathbb{F}_2}$. If $h(x) = c$, they are

$$\mathbb{F}_2^{n+1} \xrightarrow{\ p\ } \mathbb{F}_2 \xrightarrow{\ j\ } \mathbb{F}_2^{m+1}, \qquad p(x, t) = t, \qquad j(t) = (c, t).$$

Both maps admit node maps. If $h \circ f = g \circ h$, then $g(c) = c$, so the induced intermediate update is $\mathrm{Id}_{\mathbb{F}_2}$. □

If $h \circ h = h$, then $(h \otimes \mathrm{Id}_{\mathbb{F}_2}) \circ (h \otimes \mathrm{Id}_{\mathbb{F}_2}) = h \otimes \mathrm{Id}_{\mathbb{F}_2}$. Proposition 3.2.5 therefore gives a retraction onto its positive-dimensional image even when h is constant.

3.2.5 Pairings and extensions

Proposition 3.2.14 (Compatible pairing)**.** *Let $a\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ and $b\colon \mathbb{F}_2^n \to \mathbb{F}_2^s$ admit node maps. Write π_1, π_2 for the two block projections from $\mathbb{F}_2^{r+s}$. There is a map $u\colon \mathbb{F}_2^n \to \mathbb{F}_2^{r+s}$ admitting a node map with*

$$\pi_1 \circ u = a, \qquad \pi_2 \circ u = b$$

if and only if

$$S_i(a) \cap S_j(b) = \varnothing \qquad \text{for all } i \in [r],\ j \in [s].$$

When it exists, it is unique and equals $u(x) = (a(x), b(x))$. If $a \circ f = g \circ a$ and $b \circ f = k \circ b$, then $u \circ f = (g \otimes k) \circ u$.

Proof. The projection identities determine $u(x) = (a(x), b(x))$. Its components have disjoint essential supports within each block. The stated condition is therefore equivalent to all components having disjoint essential supports, hence to u admitting a node map. The intertwining identity follows by evaluation. □

For $a(t) = b(t) = t$, the pairing $u(t) = (t, t)$ admits no node map. More generally, two copies of I_1 admit no universal pairing, as the following counterexample shows.

Counterexample 3.2.15 (Failure of universal pairing)**.** Take two copies of I_1. Suppose there were a Boolean network $(\mathbb{F}_2^r, u)$ with scalar maps $\pi_1, \pi_2\colon \mathbb{F}_2^r \to \mathbb{F}_2$ satisfying $\pi_i \circ u = \pi_i$ and the following property: for every network $(\mathbb{F}_2^n, f)$ and scalar maps a, b satisfying $a \circ f = a$, $b \circ f = b$, there is a unique map $c\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ admitting a node map such that

$$c \circ f = u \circ c, \qquad \pi_1 \circ c = a, \qquad \pi_2 \circ c = b.$$

Take $f = \mathrm{Id}_{\mathbb{F}_2}$. For each constant pair $(\alpha, \beta) \in \mathbb{F}_2^2$, the required map c has the same two scalar composites as its composition with the constant-zero map on $\mathbb{F}_2$. Both intertwine the updates, so uniqueness makes c constant. Its value is a fixed state $z_{\alpha\beta}$ of u, uniquely determined by the two scalar values. These four states are distinct.

Now apply the property to the pairs $(\mathrm{Id}, 0)$, $(0, \mathrm{Id})$, and $(t \mapsto t + 1, \mathrm{Id})$. Each resulting map from $\mathbb{F}_2$ is nonconstant, hence injective. By Lemma 3.1.7, its image is a coordinate edge. The endpoint pairs are

$$\{z_{00}, z_{10}\}, \qquad \{z_{00}, z_{01}\}, \qquad \{z_{10}, z_{01}\}.$$

These pairs form a triangle, contradicting bipartiteness of the Boolean cube.

Proposition 3.2.16 (Obstruction to universal extension)**.** *Fix networks $(\mathbb{F}_2^n, f)$ and $(\mathbb{F}_2^m, g)$. There is no network $(\mathbb{F}_2^r, u)$ with maps $i\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ and $j\colon \mathbb{F}_2^m \to \mathbb{F}_2^r$ admitting node maps and satisfying $i \circ f = u \circ i$, $j \circ g = u \circ j$, with the following property: for every network $(\mathbb{F}_2^s, v)$ and maps $a\colon \mathbb{F}_2^n \to \mathbb{F}_2^s$, $b\colon \mathbb{F}_2^m \to \mathbb{F}_2^s$ admitting node maps with $a \circ f = v \circ a$, $b \circ g = v \circ b$, there is a unique map $c\colon \mathbb{F}_2^r \to \mathbb{F}_2^s$ admitting a node map such that*

$$c \circ u = v \circ c, \qquad c \circ i = a, \qquad c \circ j = b.$$

Proof. Use the constant maps with values 00 and 11 into $(\mathbb{F}_2^2, \mathrm{Id})$. The property gives $c\colon \mathbb{F}_2^r \to \mathbb{F}_2^2$. Its scalar components c_1, c_2 satisfy $c_i \circ u = c_i$ and have the same prescribed composites with both i

and j. Apply uniqueness with target $(\mathbb{F}_2, \mathrm{Id})$ to get $c_1 = c_2 = q$. This scalar function is nonconstant: it is zero on the nonempty image of i and one on the nonempty image of j. Thus $c = (q, q)$ has overlapping essential supports and cannot admit a node map. □

The same proofs establish both obstructions for Boolean maps without prescribed updates.

3.2.6 Obstructions to universal parameterization

Consider representations $H(x,t) = E(a(x),t)$ through a fixed evaluation map

$$E\colon \mathbb{F}_2^r \times \mathbb{F}_2 \to \mathbb{F}_2,$$

where a admits a node map. The following theorem gives a lower bound on the parameter dimension for a specific family of Boolean functions.

Theorem 3.2.17 (Injectivity from parity and AND)**.** *For $n \geq 1$, put*

$$H_n(x,t) = (1+t)(x_1 + \cdots + x_n) + t x_1 \cdots x_n.$$

Suppose $a\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ admits a node map and $E\colon \mathbb{F}_2^{r+1} \to \mathbb{F}_2$ satisfies

$$H_n(x,t) = E(a(x),t) \qquad (x \in \mathbb{F}_2^n,\ t \in \mathbb{F}_2).$$

Then a is injective. In particular, $r \geq n$ and a has the form in Lemma 3.1.7.

Proof. Choose a node map ϕ for a. At $t = 0$, the function H_n is parity and changes whenever one input is flipped. Thus $a(x + e_i) \neq a(x)$ for every x, i. Compatibility then gives

$$a(x + e_i) = a(x) + e_{\phi(i)}.$$

If $i \neq j$ and $\phi(i) = \phi(j)$, applying this identity twice gives

$$a(x + e_i + e_j) = a(x).$$

Take $x = \mathbf{1}$ and $t = 1$. The AND value is one at $\mathbf{1}$ and zero at $\mathbf{1} + e_i + e_j$, contradicting the assumed expression through a. Therefore ϕ is injective. Changing the nonzero inputs of x one at a time now gives

$$a(x) = a(\mathbf{0}) + \sum_{\{i : x_i = 1\}} e_{\phi(i)}.$$

The distinct assigned coordinates show that a is injective. The argument also covers $n = 1$, when injectivity of ϕ is automatic. □

Corollary 3.2.18 (No fixed universal evaluation)**.** *There is no fixed positive integer r and map $E\colon \mathbb{F}_2^{r+1} \to \mathbb{F}_2$ such that, for every $n \geq 1$ and every scalar Boolean function $H\colon \mathbb{F}_2^{n+1} \to \mathbb{F}_2$, there exists a map $a\colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ admitting a node map with*

$$H(x,t) = E(a(x),t).$$

Proof. Taking $n = r + 1$ contradicts Theorem 3.2.17, which requires $r \geq n$. A singleton parameter space also fails, since then $E(a(x),t)$ is independent of x. □

Chapter 4

Categories of Boolean Networks

4.1 The category of Boolean networks

The preceding chapter studied Boolean maps through their node maps, their fibers, and their behavior under composition. We now collect those maps into a category. The universal properties already proved will identify its quotient maps and face embeddings, and explain the role of the tensor construction.

A category consists of objects, morphisms between them, and an associative composition with an identity morphism at each object. An isomorphism is a morphism with a two-sided inverse. A functor assigns objects and morphisms while preserving identities and composition. It is faithful if it is injective on each set of morphisms, full if it is surjective on each such set, and essentially surjective if every object of its target is isomorphic to an object in its image. A full, faithful, and essentially surjective functor gives an equivalence of categories.

Definition 4.1.1. The category **BoolNet** has Boolean networks $(\mathbb{F}_2^n, f)$, with $n \geq 1$, as objects. A morphism

$$h\colon (\mathbb{F}_2^n, f) \longrightarrow (\mathbb{F}_2^m, g)$$

is a Boolean map admitting a node map and satisfying $h \circ f = g \circ h$. Composition and identities are those of the state maps.

No injectivity or surjectivity is required. A surjective morphism is a semiconjugacy under the convention of Definition 2.2.1. A morphism is determined by its state map: the choice of a compatible node map is not part of its data. We write $\mathrm{Hom}_{\mathbf{BoolNet}}(X, Y)$ for the finite set of morphisms from X to Y.

Proposition 4.1.2. *Definition 4.1.1 gives a category. Its isomorphisms are precisely the conjugacies that preserve Hamming distance.*

Proof. The identity state map admits the identity node map. Node maps compose by Proposition 3.1.2, and the identities $h \circ f = g \circ h$ and $k \circ g = r \circ k$ give

$$(k \circ h) \circ f = r \circ (k \circ h).$$

Associativity and the identity laws follow from composition of functions.

The state map of an isomorphism is bijective, so its cubes have equal dimension. By Lemma 3.1.7, it is a coordinate permutation followed by possible coordinate complementations. Its inverse has the same form and intertwines the updates in the reverse direction. Conversely, every Hamming isometry has this form by Theorem 2.1.14. A conjugacy of this form and its inverse are morphisms. □

4.1.1 Cancellation and factorization

An epimorphism permits right cancellation and a monomorphism permits left cancellation. Thus h is epic when $a \circ h = b \circ h$ forces $a = b$, and monic when $h \circ a = h \circ b$ forces $a = b$, for all morphisms with the indicated composites. A strong monomorphism is a monomorphism having the unique lifting property against every epimorphism on the left, in every commutative square with that monomorphism on the right.

Theorem 4.1.3. *The epimorphisms of* **BoolNet** *are precisely the surjective state maps. Its strong monomorphisms are precisely the injective state maps. Every nonconstant morphism has an epimorphism–strong-monomorphism factorization, unique up to an isomorphism of intermediate networks.*

Proof. Proposition 3.2.6 identifies epimorphisms with surjective state maps. An injective state map permits left cancellation, and Theorem 3.2.2 gives its lifting property against those maps. Conversely, a strong monomorphism has this lifting property, so Theorem 3.2.3 makes its state map injective.

Theorem 3.1.8 gives the asserted factorization through the invariant image face. If $h = j \circ p = j' \circ p'$ are two such factorizations, that theorem supplies a Hamming isometry u with $j' = j \circ u$ and $p' = u^{-1} \circ p$. Write $\bar{g}$ and $\bar{g}'$ for the intermediate updates, and g for the target update. Then

$$j \circ u \circ \bar{g}' = j' \circ \bar{g}' = g \circ j' = g \circ j \circ u = j \circ \bar{g} \circ u.$$

Cancelling j gives $u \circ \bar{g}' = \bar{g} \circ u$. Thus the comparison is a conjugacy, and hence an isomorphism by Proposition 4.1.2. □

Thus the first part of a nonconstant morphism records its fiber relation, and the second places the quotient dynamics on its image face. The universal properties of this image are those of Theorem 3.2.1. Up to an isomorphism of its domain, every strong monomorphism is the inclusion of an invariant coordinate face of positive dimension.

A constant morphism cannot have this factorization while node counts remain positive. In a factorization $h = j \circ p$ with p surjective and j injective, the image of j equals the image of h. A constant h would therefore require the intermediate cube to have one state.

Proposition 4.1.4. *Every nonconstant idempotent morphism splits in* **BoolNet**. *A constant idempotent does not split in* **BoolNet**.

Proof. A splitting consists of morphisms $p\colon X \to Z$ and $j\colon Z \to X$ with $h = j \circ p$ and $p \circ j = \mathrm{Id}_Z$. Proposition 3.2.5 gives these equations for a nonconstant idempotent. The second equation forces p to be surjective and j injective. If h is constant, then $\operatorname{im} j = \operatorname{im} h$ is a singleton. Injectivity of j therefore forces Z to have one state, requiring a zero-node intermediate cube. □

Proposition 4.1.5. *The category* **BoolNet** *is not balanced: a morphism can be both monic and epic without being an isomorphism. Its epimorphisms need not be regular epimorphisms.*

Proof. The majority map in Example 3.2.8 is surjective and permits left cancellation against all morphisms with the indicated target. It is therefore both epic and monic. Its two state fibers each have size four, so it is not an isomorphism.

A coequalizer of $a, b\colon Z \to X$ is a morphism $e\colon X \to Y$ with $e \circ a = e \circ b$ through which every morphism with this equality factors uniquely. Such a morphism is called a regular epimorphism. Suppose a monomorphism e is a coequalizer of a, b. From $e \circ a = e \circ b$ and monicity we obtain $a = b$. The identity of X therefore also coequalizes them, so there is $d\colon Y \to X$ with $d \circ e = \mathrm{Id}_X$. Uniqueness in the coequalizer property makes e epic, and the equality $(e \circ d) \circ e = e$ implies $e \circ d = \mathrm{Id}_Y$. Hence e is an isomorphism. The majority map above cannot be a regular epimorphism. □

4.1.2 Tensor and universal constructions

We use the tensor of Definition 3.2.9 and retain the notation $I_n = (\mathbb{F}_2^n, \mathrm{Id}_{\mathbb{F}_2^n})$ for the identity networks.

For this two-argument construction, functoriality means that pairs of identity maps give an identity and that tensoring satisfies the interchange identity for composition. This is the meaning of bifunctor in the next proposition. The associativity and symmetry identifications are coherent when any two sequences of regrouping and exchanging blocks that give the same final permutation of the blocks induce the same map.

Proposition 4.1.6. *The tensor is a bifunctor on* **BoolNet**. *Regrouping coordinate blocks and exchanging them give coherent associativity and symmetry isomorphisms. It preserves epimorphisms and strong monomorphisms, and carries image factorizations of nonconstant morphisms to image factorizations.*

Proof. Proposition 3.2.10 gives compatibility with composition and identities, and shows that tensoring preserves surjectivity and injectivity. Apply Theorem 4.1.3. The associativity and symmetry maps are the coordinate permutations that regroup or exchange the ordered blocks; their coherence identities follow by tracking each block. Theorem 3.2.11 gives the assertion about image factorizations. □

Theorem 3.2.13 also shows that $h \otimes \mathrm{Id}_{I_1}$ has an epimorphism–strong-monomorphism factorization for every morphism h, including constants. Every idempotent becomes split after tensoring with Id_{I_1}.

A product of X and Y is a network P with projections to X and Y through which every pair of morphisms from a common source factors uniquely. A coproduct has the reverse property: it receives maps from X and Y, and every pair of morphisms from them to a common target extends uniquely. These are exactly the pairing and extension properties studied in Section 3.2.5. An initial object has exactly one morphism to each object, and a terminal object has exactly one morphism from each object. A tensor unit would have natural isomorphisms $X \otimes U \cong X \cong U \otimes X$ compatible with regrouping.

Proposition 4.1.7. *The product of I_1 with itself does not exist in* **BoolNet**. *No pair of objects has a coproduct. There is no initial object, no terminal object, and no tensor unit.*

Proof. The universal properties of a product and a coproduct are exactly the pairing and extension properties excluded by Counterexample 3.2.15 and Proposition 3.2.16, respectively. In particular, the state projections of a tensor do not make it a categorical product; Proposition 3.2.14 specifies when a pairing admits a node map.

Every network has two distinct constant morphisms to I_1, so no object is initial. If T were terminal, its unique incoming morphism from I_1 would take values in the fixed states of T. Choose such a state c. The identity of T and the constant endomorphism at c would be distinct, contrary to terminality.

If a tensor unit had $r \geq 1$ nodes, its tensor with I_1 would have 2^{r+1} states. It could not be isomorphic to I_1, which has two states. □

4.1.3 The singleton and splitting idempotents

The missing tensor unit and the unsplit constant idempotents have the same source: the singleton network has not been included. We now make this extension explicit.

Definition 4.1.8. Let $I_0 = (\mathbb{F}_2^0, \mathrm{Id}_{\mathbb{F}_2^0})$, where $\mathbb{F}_2^0$ consists of the empty tuple. The category $\mathbf{BoolNet}_0$ has the objects of **BoolNet** together with I_0. Between objects of **BoolNet**, its morphisms are unchanged. There is one morphism $X \to I_0$ for every object X, and a morphism $I_0 \to (\mathbb{F}_2^n, f)$

is a state $c \in \mathbb{F}_2^n$ with $f(c) = c$, regarded as a constant state map. Composition and identities are those of the state maps. Tensor is given by the same coordinate concatenation as in Definition 3.2.9, now allowing empty coordinate blocks.

The unique endomorphism of I_0 is its identity. These rules are closed under composition: a morphism carries a fixed state to a fixed state, and a composite through I_0 is constant at a fixed state of its target. This defines a category containing **BoolNet** as a full subcategory. Permitting maps into I_0 explicitly extends the node map convention, since there is no total node map $[n] \to [0]$ when $n > 0$.

Tensor also respects these rules. For positive-dimensional source and target, its output components have disjoint essential supports, just as in Proposition 3.2.10; an empty output block discards its inputs, and an empty input block contributes constant coordinates. If the entire source is I_0, the tensor value is a fixed state; if the entire target is I_0, the map is the unique one. The interchange identity and the identities for updates hold for empty blocks as well. Thus $\mathbf{BoolNet}_0$ is symmetric monoidal, with unit I_0 and the coordinate identifications as unit isomorphisms. The singleton is also terminal. Every constant morphism factors through it.

This extension can be identified with a standard categorical construction. The *Karoubi envelope* $\mathrm{Kar}(\mathbf{BoolNet})$ has pairs (X, e) as objects, where $e\colon X \to X$ is a morphism satisfying $e \circ e = e$. A morphism

$$a\colon (X, e) \longrightarrow (Y, d)$$

is a morphism $a\colon X \to Y$ in **BoolNet** satisfying

$$a = d \circ a \circ e, \qquad \text{equivalently} \qquad d \circ a = a = a \circ e.$$

Composition is inherited from **BoolNet**, and the identity of (X, e) is e. The displayed equations ensure that composition is closed and that these are identity morphisms.

Proposition 4.1.9 (Splitting all idempotents)**.** *Every idempotent in* $\mathbf{BoolNet}_0$ *splits, and there is an equivalence*

$$\mathrm{Kar}(\mathbf{BoolNet}) \simeq \mathbf{BoolNet}_0.$$

Proof. For each idempotent $e\colon X \to X$ in **BoolNet**, choose maps in $\mathbf{BoolNet}_0$

$$X \xrightarrow{p_e} Z_e \xrightarrow{j_e} X, \qquad j_e \circ p_e = e, \qquad p_e \circ j_e = \mathrm{Id}_{Z_e}.$$

If e is nonconstant, Proposition 3.2.5 supplies such a splitting through its image face. If e is constant at c, its intertwining identity makes c a fixed state of X. Take $Z_e = I_0$, let p_e be the unique map to I_0, and let j_e select c. For $e = \mathrm{Id}_X$, choose $Z_e = X$ and both maps to be identities. This covers every idempotent on a positive-dimensional object. The only idempotent on I_0 is its identity, which already splits.

Define $K\colon \mathrm{Kar}(\mathbf{BoolNet}) \to \mathbf{BoolNet}_0$ by

$$K(X, e) = Z_e, \qquad K(a) = p_d \circ a \circ j_e$$

for $a\colon (X, e) \to (Y, d)$. It sends the identity e to Id_{Z_e}. If also $b\colon (Y, d) \to (W, k)$, then

$$K(b) \circ K(a) = p_k \circ b \circ j_d \circ p_d \circ a \circ j_e = p_k \circ b \circ d \circ a \circ j_e = p_k \circ b \circ a \circ j_e = K(b \circ a).$$

Thus K is a functor.

For a morphism $v\colon Z_e \to Z_d$ in $\mathbf{BoolNet}_0$, put

$$a = j_d \circ v \circ p_e.$$

Its source and target lie in **BoolNet**, so fullness of the inclusion makes it a morphism of **BoolNet**. The splitting identities give $d \circ a \circ e = a$, so it is a morphism in the Karoubi envelope. Moreover,

$$p_d \circ (j_d \circ v \circ p_e) \circ j_e = v, \qquad j_d \circ (p_d \circ a \circ j_e) \circ p_e = d \circ a \circ e = a.$$

These formulas are inverse on the corresponding hom-sets, proving that K is full and faithful. They apply also when one or both of Z_e, Z_d equal I_0.

Every object X of **BoolNet** is $K(X, \mathrm{Id}_X)$. The constant-zero endomorphism e_0 of I_1 is idempotent and has $K(I_1, e_0) = I_0$. Hence K is essentially surjective, and therefore an equivalence. □

Adjoining the singleton therefore realizes the idempotent completion of **BoolNet**. The product obstruction for I_1 and the coproduct obstruction persist. For positive-dimensional candidate objects, the same proofs apply, including when a source of a proposed coproduct is I_0: each source is nonempty and has constant maps to the identity networks. A singleton candidate cannot realize the distinct projection values required for the product, or the simultaneous values 00 and 11 required for a coproduct.

4.1.4 The obstruction to an internal hom

A right adjoint R to a functor F gives bijections $\mathrm{Hom}(F(X), Y) \cong \mathrm{Hom}(X, R(Y))$ natural in X and Y. In particular, there is a morphism $\varepsilon_Y \colon F(R(Y)) \to Y$, called the counit, such that every $q \colon F(X) \to Y$ has a unique expression $q = \varepsilon_Y \circ F(a)$. For $F = - \otimes A$, the object $R(Y)$ is an internal hom from A to Y. A symmetric monoidal category is closed if these right adjoints exist for every A.

Theorem 4.1.10. *The functor $- \otimes I_1 \colon$ **BoolNet** $\to$ **BoolNet** has no right adjoint.*

Proof. Suppose that it has a right adjoint R, and put $H = R(I_1) = (\mathbb{F}_2^r, u)$. The counit at I_1 is a morphism

$$E \colon H \otimes I_1 \longrightarrow I_1.$$

For every $n \geq 1$ and every morphism $q \colon I_n \otimes I_1 \to I_1$, the adjunction supplies a morphism $a \colon I_n \to H$ satisfying

$$q = E \circ (a \otimes \mathrm{Id}_{I_1}).$$

Every scalar Boolean function on $\mathbb{F}_2^{n+1}$ is such a morphism, since both updates are identities. Thus a single Boolean map $E \colon \mathbb{F}_2^{r+1} \to \mathbb{F}_2$ would evaluate every one of these functions through a map $a \colon \mathbb{F}_2^n \to \mathbb{F}_2^r$ admitting a node map. Corollary 3.2.18 excludes this even without requiring the update identity for a. Hence no right adjoint exists. □

The argument also applies in $\mathbf{BoolNet}_0$. If $H = I_0$, the expression $E(a(x), t)$ is independent of x and cannot equal the projection $(x, t) \mapsto x_1$. When H has positive dimension, the preceding proof applies. Consequently, $\mathbf{BoolNet}_0$ is not closed: there is no internal hom for the pair (I_1, I_1).

4.2 Comparison with finite dynamical systems

Let **FDS** denote the category whose objects are finite sets S with a map $u \colon S \to S$, and whose morphisms $h \colon (S, u) \to (T, v)$ are functions satisfying $h \circ u = v \circ h$. This is the familiar category of finite actions of the additive monoid of nonnegative integers. Indeed, an update u gives the action $k \cdot s = u^k(s)$, with $0 \cdot s = s$ and $(k + \ell) \cdot s = k \cdot (\ell \cdot s)$. Conversely, an action is determined by the update $u(s) = 1 \cdot s$. A function commutes with this update if and only if it commutes with every iterate, which is exactly equivariance of the corresponding actions.

Proposition 4.2.1. *Sending $(\mathbb{F}_2^n, f)$ to its finite state set with update f, and each morphism to its state map, defines a faithful functor*

$$U\colon \mathbf{BoolNet} \longrightarrow \mathbf{FDS}.$$

It reflects isomorphisms but is not full. Up to isomorphism, its image on objects consists precisely of finite dynamical systems with 2^n states for some $n \geq 1$. It carries tensor to the Cartesian product of finite dynamical systems. Neither **BoolNet** *nor* $\mathbf{BoolNet}_0$ *is equivalent to* **FDS**.

Proof. Identities and composition are unchanged, so U is a functor. It is faithful because a morphism of **BoolNet** is determined by its state map. If $U(h)$ is an isomorphism, then h is bijective. Its source and target have the same dimension, and Lemma 3.1.7 makes it a coordinate permutation followed by coordinate complementations. Its inverse admits a node map and intertwines the updates. Thus h is already an isomorphism in **BoolNet**.

The map $t \mapsto (t, t)$ from $U(I_1)$ to $U(I_2)$ is a morphism in **FDS**, since both updates are identities. It has no node map, so it is not the image of a morphism in **BoolNet**. Hence U is not full.

Every object in the image has 2^n states. Conversely, if $|S| = 2^n$ with $n \geq 1$, choose a bijection $q\colon \mathbb{F}_2^n \to S$. The Boolean network with update $q^{-1} \circ u \circ q$ maps to an object of **FDS** isomorphic to (S, u) through q. This proves the assertion about objects; for example, the three-state identity system is not in this essential image.

The product of (S, u) and (T, v) in **FDS** has state set $S \times T$ and update $(s, t) \mapsto (u(s), v(t))$. Its projections commute with updates, and the unique pairing of two commuting functions also commutes with updates. Under the identification $\mathbb{F}_2^{n+m} = \mathbb{F}_2^n \times \mathbb{F}_2^m$, this is exactly the image under U of the tensor construction.

The singleton identity system is terminal in **FDS**, whereas **BoolNet** has no terminal object by Proposition 4.1.7. Existence of a terminal object is preserved by an equivalence, so no equivalence between these categories is possible. The category **FDS** has the product of every pair of objects, while $\mathbf{BoolNet}_0$ has no product of I_1 with itself. This also rules out an equivalence between $\mathbf{BoolNet}_0$ and **FDS**. □

Thus actions on finite state sets describe the underlying dynamics. The additional restriction on coordinate dependence is recorded by the morphisms of **BoolNet**. The equivalence with a standard completion established here is Proposition 4.1.9: adjoining the singleton realizes the Karoubi envelope of **BoolNet**.

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